

CHAPTER TWENTY

The Evolution of High Visual Acuity in the Anthroidea

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INTRODUCTION

Most definitions of the order primates make some reference to the importance of vision relative to the other special senses (Le Gros Clark, 1959; Martin, 1990; Napier and Napier, 1967). This characterization is particularly fitting for haplorhine primates, in which vision is unquestionably the dominant sensory modality. Anthropoids and tarsiers differ from strepsirrhines in exhibiting a derived reduction of the olfactory apparatus (Baron et al., 1983; Cave, 1973) and an elaboration of the visual sense to a degree that is unique among mammals (Rodieck, 1973; Walls, 1942). Visual adaptations, however, are divergent within the haplorhine suborder. While the tarsier visual system is primarily specialized for enhanced sensitivity in the context of nocturnal visual predation (Castenholz, 1984), most living anthropoids are adapted for extremely acute diurnal vision.

Visual acuity, or resolving power, refers to an ability to distinguish between closely-spaced visual stimuli. Where the optical properties of the eye do not limit acuity (e.g., through refractive errors or large pupillary diameter), visual

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acuity refers to the ability of the brain to distinguish fine spatial details of the image projected by the cornea and lens onto the retina. Behavioral tests of visual acuity in anthropoids consistently demonstrate that diurnal anthropoids have a maximum acuity between about 40 and 80 c/deg¹ (Table 1). These measurements for diurnal anthropoids are substantially higher than the behaviorally-determined acuity of all other mammals (range 0.05–23.3 c/deg; Table 1), and are congruent with anatomical estimates of visual acuity based on ganglion cell density and eye size (Table 2). The visual acuity of diurnal anthropoids even exceeds known values for many diurnal birds, such as pigeons (*Columba livia*; 16 c/deg), northern blue jays (*Cyanocitta cristata*; 15–27 c/deg), and ostriches (*Struthio camelus*; 17–23 c/deg) (Boire et al., 2001; Fite and Rosenfield-Wessels, 1975; Hodos et al., 1991). The only vertebrates known to surpass diurnal anthropoids in visual acuity are large diurnal raptors such as the wedge-tailed eagle (*Aquila audax*; 140 c/deg), which have much larger eyes and higher foveal photoreceptor densities than diurnal anthropoids² (Fite and Rosenfield-Wessels, 1975; Reymond, 1985; Troilo et al., 1993; Wikler and Rakic, 1990; Yamada et al., 2001).

As the only nocturnal anthropoid, the owl monkey (*Aotus*) provides an exception to the general anthropoid pattern of specialization for acute diurnal vision. Behavioral tests of visual acuity in *Aotus* under varying levels of illumination indicate that the maximum acuity of this genus is about 10 c/deg (Jacobs, 1977). While this value for *Aotus* is significantly lower than other anthropoids, it is higher than most other mammals for which behavioral tests

¹ Early behavioral tests of visual acuity in primates expressed maximum visual acuity in terms of the retinal angular separation (in minutes, degrees, and seconds) of the closest two visual stimuli (e.g., parallel lines) that a test subject could perceive as separate entities. Many subsequent psychophysical studies of visual acuity and spatial contrast sensitivity have used a “grating” pattern as the primary test stimulus (e.g., Jacobs, 1977). A grating is simply an alternating sequence of parallel light and dark bands, often generated by an oscilloscope or computer (Woodhouse and Barlow, 1982). Grating bands may be sharply distinct stripes (e.g., a square wave grating) or may grade from light to dark regions according to a specified mathematical formula (e.g., a sinusoidally-modulated grating). Grating acuity is thus quantified by convention as the maximum number of full cycles of the grating stimulus per degree of visual angle (cycles per degree, or c/deg) that can be distinguished from a uniform gray field of equal total luminosity.

² When diurnal anthropoids are compared with smaller raptors of similar eye size, the differences in visual acuity are minimal. For instance, humans (axial eye length: 24 mm) and the Australian brown falcon *Falco berigora* (axial eye length: 23 mm) have maximum visual acuities of 77 c/deg and 73 c/deg, respectively (Cavonius and Robbins, 1973; Reymond, 1985, 1987). Anatomical estimates of acuity for raptors with smaller eyes (e.g., the American kestrel, *Falco sparverius*—46 c/deg; Dvorak et al., 1983; Hirsch, 1983) fall below these values, but within the range of acuities demonstrated for smaller-eyed anthropoids (Tables 1 and 2).

Table 1. Behavioral measurements of visual acuity in various mammalian taxa

Scientific name	Common name	Activity pattern	Visual acuity (c/deg)	Source
<i>Homo sapiens</i>	Human	Diurnal	50.0–77.0	Cavonius and Robbins, 1973; DeValois et al., 1974
<i>Macaca mulatta</i>	Rhesus macaque	Diurnal	46.2–53.6	Cowey and Ellis, 1967; Cavonius and Robbins, 1973
<i>Macaca nemestrina</i>	Pig-tailed macaque	Diurnal	46.0	DeValois et al., 1974
<i>Macaca fascicularis</i>	Crab-eating macaque	Diurnal	46.0	DeValois et al., 1974
<i>Saimiri sciureus</i>	Squirrel monkey	Diurnal	40.5	Cowey and Ellis, 1967
<i>Equus caballus</i>	Domestic horse	Cathemeral	10.9–23.3	Timney and Keil, 1992
<i>Aotus trivirgatus</i>	Northern owl monkey	Nocturnal	10.0	Jacobs, 1977
<i>Camelus bactrianus</i>	Bactrian camel	Cathemeral	10.0	Harman et al., 2001
<i>Felis catus</i>	Domestic cat	Cathemeral	5.0–8.9	Bisti and Maffei, 1974; Blake et al., 1974; Jacobson et al., 1976
<i>Lemur catta</i>	Ring-tailed lemur	Diurnal	6.0–6.7	Neuringer et al., 1981
<i>Suricata suricatta</i>	Meerkat	Diurnal	6.3	Moran et al., 1983
<i>Otolemur crassicaudatus</i>	Fat-tailed bush baby	Nocturnal	4.8–6.0	Langston et al., 1986
<i>Zalophus californicus</i>	California sea lion	Cathemeral	5.4–5.7	Schusterman and Balliet, 1970b
<i>Myrmecobius fasciatus</i>	Numbat	Diurnal	5.2	Arrese et al., 2000
<i>Lagenorhynchus obliquidens</i>	Pacific white-sided dolphin	Cathemeral	5.0	Spong and White, 1971
<i>Eumetopias jubata</i>	Stellar sea lion	Cathemeral	4.2 (water)	Schusterman and Balliet, 1970a
<i>Spermophilus beecheyi</i>	California ground squirrel	Diurnal	4.0	Jacobs et al., 1980
<i>Sciurus griseus</i>	Western gray squirrel	Diurnal	3.9	Jacobs et al., 1982
<i>Sciurus niger</i>	Fox squirrel	Diurnal	3.9	Jacobs et al., 1982
<i>Sciurus carolinensis</i>	Eastern gray squirrel	Diurnal	3.9	Jacobs et al., 1982
<i>Tursiops truncatus</i>	Bottlenose dolphin	Cathemeral	1.6–2.5 (air); 2.5–3.8 (water)	Herman et al., 1975

Table 1. Continued

Scientific name	Common name	Activity pattern	Visual acuity (c/deg)	Source
<i>Phoca vitulina</i>	Harbour seal	Cathemeral	3.6	Schusterman and Balliet, 1970a
<i>Pteropus giganteus</i>	Indian flying fox	Nocturnal	3.5	Neuweiler, 1962; cited in Pettigrew et al., 1988
<i>Oryctolagus cuniculus</i>	Old World rabbit	Cathemeral	1.5–3.0	Van Hof, 1967; Vaney, 1980
<i>Dasyurus hallucatus</i>	Quoll	Nocturnal	2.3–2.8	Harman et al., 1986
<i>Amblonyx cinerea</i>	Asian clawless otter	Cathemeral	1.9–2.2 (air); 2.0 (water)	Balliet and Schusterman, 1971
<i>Sminthopsis crassicaudata</i>	Fat-tailed dunnart	Nocturnal	2.4	Arrese et al., 1999
<i>Tupaia belangeri</i>	Tree shrew	Diurnal	1.2–2.4	Petry et al., 1984
<i>Mustela putorius</i>	European polecat	Cathemeral	1.2–1.9	Neumann and Schmidt, 1959, cited in Mass and Supin, 2000
<i>Mustela erminea</i>	Ermine	Cathemeral	1.2–1.9	Neumann and Schmidt, 1959, cited in Mass and Supin, 2000
<i>Rattus norvegicus</i>	Pigmented laboratory rat	Nocturnal	1.2–1.6	Birch and Jacobs, 1979; Seymour and Juraska, 1997
<i>Tarsipes rostratus</i>	Honey possum	Cathemeral	0.6	Arrese et al., 2002
<i>Mus musculus</i>	House mouse	Cathemeral	0.5	Sinex et al., 1979; Prusky et al., 2000
<i>Mesocricetus auratus</i>	Golden hamster	Nocturnal	0.5	Emerson, 1980
<i>Phyllostomus hastatus</i>	Greater spear-nosed bat	Nocturnal	0.1–0.4	Suthers, 1966
<i>Artibeus jamaicensis</i>	Jamaican fruit-eating bat	Nocturnal	0.1–0.4	Suthers, 1966
<i>Myotis lucifugus</i>	Little brown bat	Nocturnal	0.05–0.1	Suthers, 1966

Notes: Visual acuity is expressed in cycles per degree, and reflects the highest acuity attainable using established behavioral testing protocols (primarily conditioned responses to grating stimuli). All tests were undertaken in air (versus under water) unless otherwise specified. Activity pattern follows the cited source, or sources cited in Nowak (1991) and Rowe (1996). Species that are active exclusively or nearly exclusively during the day or night are classified as diurnal and nocturnal, respectively. Species that are active throughout a 24-h cycle, or have multiple peaks of activity that occur at both day and night have been classified as cathemeral.

Table 2. Anatomical estimates of visual acuity in various mammalian taxa

Scientific name	Common name	Activity pattern	Visual acuity (c/deg)	Source
<i>Homo sapiens</i>	Human	Diurnal	47.0–86.0	Hirsch and Curcio, 1989; Curcio et al., 1990
<i>Cebus apella</i>	Tufted capuchin	Diurnal	38.8–54.8	Andrade da Costa and Hokoç, 2000
<i>Callithrix jacchus</i>	Common marmoset	Diurnal	30.0	Troilo et al., 1993
<i>Equus caballus</i>	Domestic horse	Cathemeral	16.4–20.4	Timney and Keil, 1992
<i>Camelus dromedarius</i>	Dromedary camel	Cathemeral	10.4	Harman et al., 2001
<i>Aotus azarae</i>	Azara's owl monkey	Nocturnal or Cathemeral	8.3	Yamada et al., 2001
<i>Felis catus</i>	Domestic cat	Cathemeral	8.1	Hughes, 1975
<i>Otolemur crassicaudatus</i>	Fat-tailed bush baby	Nocturnal	6.2–7.5	DeBruyn et al., 1980; Dkhissi-Benyahya et al., 2001
<i>Callorhinus ursinus</i>	Northern fur seal	Cathemeral	5.6–7.1	Mass and Supin, 1992
<i>Myrmecobius fasciatus</i>	Numbat	Diurnal	6.3	Arrese et al., 2000
<i>Pteropus poliocephalus</i>	Gray-headed flying fox	Nocturnal	5.5	Pettigrew et al., 1988
<i>Microcebus murinus</i>	Gray mouse lemur	Nocturnal	4.9	Dkhissi-Benyahya et al., 2001
<i>Trichosurus vulpecula</i>	Brush-tailed possum	Nocturnal	4.8	Freeman and Tancred, 1978
<i>Enhydra lutris</i>	Sea otter	Diurnal	4.2 (air and water)	Mass and Supin, 2000
<i>Balaenoptera acutorostrata</i>	Minke whale	Cathemeral	3.9–4.2 (water)	Murayama et al., 1992
<i>Loxodonta africana</i>	African elephant	Cathemeral	4.1	Stone and Halasz, 1989
<i>Pteropus scapulatus</i>	Little red flying fox	Nocturnal	4.0	Pettigrew et al., 1988
<i>Odobenus rosmarus</i>	Pacific walrus	Cathemeral	3.8	Mass, 1992
<i>Delphinus delphis</i>	Common dolphin	Cathemeral	3.8 (water)	Dral, 1983
<i>Pseudorca crassidens</i>	False killer whale	Cathemeral	3.3 (water)	Murayama and Somiya, 1998
<i>Tursiops truncatus</i>	Bottlenose dolphin	Cathemeral	2.5 (air); 3.3 (water)	Mass and Supin, 1995
<i>Rousettus</i> sp.	Rousette fruit bat	Nocturnal	3.0	Marks, 1980; cited in Pettigrew et al., 1988
<i>Phocoena phocoena</i>	Harbor porpoise	Cathemeral	2.1 (air); 2.7 (water)	Mass and Supin, 1986
<i>Eschrichtius robustus</i>	Gray whale	Cathemeral	2.7 (water)	Mass and Supin, 1997
<i>Lagenorhynchus obliquidens</i>	Pacific white-sided dolphin	Cathemeral	2.7 (water)	Murayama and Somiya, 1998

Table 2. Continued

Scientific name	Common name	Activity pattern	Visual acuity (c/deg)	Source
<i>Delphinapterus leucas</i>	Beluga whale	Cathemeral	2.6 (water)	Murayama and Somiya, 1998
<i>Dasyurus hallucatus</i>	Quoll	Nocturnal	2.6	Harman et al., 1986
<i>Phocoenoides dalli</i>	Dall's porpoise	Cathemeral	2.5–2.6 (water)	Murayama et al., 1992, 1995
<i>Phascolarctos cinereus</i>	Koala	Cathemeral	2.4	Schmid et al., 1992
<i>Sminthopsis crassicaudata</i>	Fat-tailed dunnart	Nocturnal	2.3	Arrese et al., 1999
<i>Macroderma gigas</i>	Australian false vampire bat	Nocturnal	1.9	Pettigrew et al., 1988
<i>Mesocricetus auratus</i>	Golden hamster	Nocturnal	1.8	Tiao and Blakemore, 1976
<i>Megaderma lyra</i>	Greater false vampire bat	Nocturnal	1.5	Pettigrew et al., 1988
<i>Artibeus cinereus</i>	Gervais fruit eating bat	Nocturnal	1.4	Pettigrew et al., 1988
<i>Taphozous georgianus</i>	Brown sheath-tailed bat	Nocturnal	1.3	Pettigrew et al., 1988
<i>Sotalia fluviatilis</i>	River dolphin (tucuxi)	Cathemeral	0.9 (air); 1.2 (water)	Mass and Supin, 1999
<i>Saccopteryx</i> sp.	White-lined bat	Nocturnal	1.0	Marks, 1980; cited in Pettigrew et al., 1988
<i>Tarsipes rostratus</i>	Honey possum	Cathemeral	0.8	Dunlop et al., 1994
<i>Inia geoffrensis</i>	Amazon dolphin (botos)	Cathemeral	0.6 (air); 0.8 (water)	Mass and Supin, 1989, 1999
<i>Eptesicus</i> sp.	Big brown bat	Nocturnal	0.7	Marks, 1980; cited in Pettigrew et al., 1988
<i>Nyctophilus gouldi</i>	Gould's long-eared bat	Nocturnal	0.6	Pettigrew et al., 1988
<i>Rhinolophus rouxi</i>	Horseshoe bat	Nocturnal	0.4	Pettigrew et al., 1988

Notes: Visual acuity is expressed in cycles per degree, and reflects the highest acuity theoretically attainable based upon the morphology of the eye and the microanatomy of the retina. All calculations were made for vision in air (versus under water) unless otherwise specified. Note that the methods used by different authors to calculate anatomical acuity may vary slightly—see cited references for the specific procedure used with each species. Activity pattern source and definitions are the same as for Table 1.

of visual acuity have been undertaken (Table 1). Indeed, the only mammals currently known to have visual acuity comparable to or higher than *Aotus* are horses (*Equus*), cats (*Felis*), and camels (*Camelus*). Comparable or higher visual acuity in each of these nonprimate genera can be partly attributed to the fact that they have larger eyes than owl monkeys (axial lengths in mm: Owl monkey—16; cat—22; camel—33; horse—40; Harman et al., 2001; Ogden, 1994; Timney and Keil, 1992; Vakkur and Bishop, 1963). However, these data on eye size and acuity further serve to emphasize the extreme visual specialization of diurnal anthropoids, in which even absolutely small-eyed anthropoids such as *Callithrix jacchus* (axial length: 11 mm; Troilo et al., 1993) demonstrate acuities that surpass much larger-eyed non-anthropoid species like the horse.

Diurnality was likely a prerequisite for the evolution of anthropoid high visual acuity, because many of the sensitivity-enhancing features typical of nocturnally-adapted eyes are detrimental to visual acuity (Walls, 1942). However, the available comparative data on visual acuity in mammals (Tables 1 and 2) clearly indicate that ecological factors other than activity pattern must also have played a significant role in selecting for the unusually high visual acuity of anthropoids. Indeed, most diurnal mammals (e.g., numbats [*Myrmecobius*], treeshrews [*Tupaia*], meerkats [*Suricata*], and squirrels [*Spermophilus*, *Sciurus*]; Tables 1 and 2) demonstrate maximum visual acuities considerably lower than those of diurnal anthropoids. Comparative studies of retinal anatomy suggest that one important acuity-enhancing feature of the diurnal anthropoid eye (the retinal fovea, see section on Central Retinal Anatomy) is also typically well-developed in non-mammalian vertebrates that rely on visually-directed predation (although this association is not exclusive; Fite and Rosenfield-Wessels, 1975; Walls, 1942; Ross, this volume). This evidence lends support to the hypothesis that the acuity-enhancing visual adaptations of diurnal anthropoids are inherited from an ancestor that was a diurnal visual predator (Ross, 1996).

THE MORPHOLOGICAL BASIS FOR HIGH VISUAL ACUITY IN ANTHROPOIDS

The unusually high visual acuity of anthropoids relative to other mammals is primarily the result of several specialized features of the anthropoid peripheral visual system. These adaptations include: (a) Relatively large eye size, (b) absence of a tapetum lucidum, (c) short-wavelength filters in the lens and retina that limit chromatic aberration, and (d) central retinas with high photoreceptor densities and low retinal summation.

Eye Size and Shape

Anthropoids, like all primates, have relatively large eyes compared to other mammals (Hughes, 1977; Ritland, 1982; Ross, 2000; Schultz, 1940; Stephan et al., 1984; Walls, 1942). Larger eyes can accommodate a larger retinal photoreceptor array, and may thus support higher visual acuity by spreading the retinal image over a larger number of sensory cells (cf. Figure 5 in Ross, 2000). If an analogy is made with a computer monitor, increasing the size of the retina and retinal image (while holding photoreceptor size relatively constant) is roughly equivalent to viewing a larger monitor with a greater total number of pixels on its screen.

Ross (2000) has further shown that the eyes of diurnal anthropoids have distinctive proportions. Specifically, the ratio of the size of the cornea to the estimated posterior nodal distance³ is much lower in diurnal anthropoids than in most other mammals, including diurnal and nocturnal strepsirrhines (cf. Figures 6 and 7 in Ross, 2000). While this configuration improves visual acuity by producing a relatively large retinal image, it is detrimental to visual sensitivity because the enlarged retinal image is reduced in brightness. Indeed, Ross (2000) reports that under comparable lighting conditions, the brightness of the retinal image is approximately five times less in diurnal anthropoids than in other mammals. Furthermore, small corneal size limits the maximum possible light-gathering power of the anthropoid eye by constraining the maximum effective diameter of the pupil. These features suggest that the eyes of diurnal anthropoids are specifically adapted to sustain high acuity in a setting where high ambient light levels compensate for the proportional dimness of the retinal image.

Absence of Tapeta Lucida

The tapetum lucidum (literally “bright carpet”) is a layer of reflective tissue that lies behind the photosensitive cells of the retina. In a tapetum-bearing eye, much

³ Of the light rays entering a vertebrate eye, a subset is refracted by the lens and cornea in such a way that they travel along an emergent path within the eye that is roughly parallel to the incident path outside the eye (McIlwain, 1996). The point from which these light rays diverge to spread across the retina is referred to as the posterior nodal point, and is typically located within the lens along its central axis. The posterior nodal distance (PND) is the distance between the retina and the posterior nodal point, and is thus largely a function of eye length and refractive power. PND is an important determinant of retinal image size, and is usually calculated for diurnal anthropoids as the product of the eye's axial length and the constant 0.63 (Hughes, 1977; Ross, 2000).

of the light that passes through the retina without being absorbed is reflected by the tapetum back toward the photoreceptor layer. Tapeta thus greatly increase the likelihood that a single photon of light will be captured by the retinal photoreceptor array. However, tapeta also unavoidably decrease visual acuity by scattering light within the eye (Buttery et al., 1990; Walls, 1942).

All anthropoids and tarsiers lack a tapetum lucidum⁴ (Martin, 1990). By contrast, tapeta are widespread among mammals, and appear to have arisen convergently in most mammalian orders (Nicol, 1981; Pirie, 1966). With the exception of the genera *Eulemur* and *Varecia*, which lack tapeta, strepsirrhine primates are unique among mammals in possessing choroidal tapeta composed of crystalline riboflavin (Alfieri et al., 1976; Dartnall et al., 1965; Martin, 1990; Pariente, 1976; Pirie, 1959; Rohen and Castenholz, 1967; Wolin and Massopust, 1970). While the absence of a tapetum may indeed be a synapomorphy of the Haplorhini (Cartmill, 1980; Martin, 1973; Ross et al., 1998), the polarity of this character is uncertain. If extant species alone are considered, it is equally parsimonious to assume that a riboflavin-based tapetum is a synapomorphy of strepsirrhines, and that its absence in the Haplorhini is a primitive retention from the last common ancestor of living primates.

Although diurnality is typically regarded as the root cause of tapetal absence in living anthropoids (Martin, 1990), comparative evidence argues against a simple cause and effect relationship with activity pattern. Indeed, not all nocturnal mammals possess tapeta (e.g., *Tarsius*), and not all diurnal species lack them (e.g., *Propithecus* and *Lemur*). In particular, it is noteworthy that many mammals that are regularly active in bright sunlight retain tapeta, including many diurnal strepsirrhines, cathemeral carnivorans, and cathemeral ungulates

⁴ In a review of the comparative literature describing primate visual anatomy, it is not uncommon to encounter statements affirming the presence of a tapetum in owl monkeys (e.g., Rodieck, 1988, p. 206). Most such claims can likely be attributed to Walls' initial report of an "excellent tapetum fibrosum" in *Aotus* (Walls, 1942, p. 233). However, Walls subsequently retracted this statement, noting that "In *The Vertebrate Eye* I erred in ascribing a tapetum lucidum (fibrosum) to *Aotus*... I do not now know the source of this mistake, nor of my information that *Aotus* has vivid eyeshine" (Walls, 1953, p. 62). Indeed, histological studies of owl monkey retinas have invariably failed to identify the presence of a tapetum (Detwiler, 1943; Kolmer, 1930; Rohen and Castenholz, 1967). Whatever the origin of Walls' original claim, it is likely due to the fact that wild populations of *Aotus* do have conspicuous eyeshine (Thorington et al., 1975). Similarly, *in vivo* studies of owl monkey retinas have confirmed the presence of a reflective fundus (Ogden, 1983). Rather than being the result of tapetal reflection, however, it is probable that eyeshine in *Aotus* is the result of myelination of the inner plexiform and nerve fiber layers of the retina (Jones, 1965). This myelination of the nerve fiber layer causes fascicles of ganglion cell axons on the inner retinal surface to have the appearance of "reflective wedge-shaped spokes radiating from the optic disc" when viewed funduscopically (Ogden, 1994, p. 269).

(Martin, 1990; Nicol, 1981; Pirie, 1966). In such species, the tapetum likely places an upper limit on visual acuity because the resolving power of the photoreceptor mosaic cannot be increased beyond the threshold set by intraocular tapetal scattering. Anthropoids and tarsiers, by contrast, lack tapeta entirely, and thus face no obstacle to increasing visual acuity up to the threshold dictated by the properties of their optical media and the light levels of their environments.

Short-Wavelength Filters

The degree to which light is refracted by the eye's cornea and lens is an inverse function of its wavelength. Short-wavelength light in the blue and ultraviolet (UV) part of the spectrum is refracted more than light of longer wavelengths, and thus short-wavelength light comes into sharp focus in front of (rather than on) the retina of an eye that is free of refractive errors (Walls, 1942). Blue and violet images thus appear slightly blurred relative to red and yellow images even when the eye is in focus. This problem of "chromatic aberration" is minimized in the eyes of diurnal anthropoids by the presence of two types of optical filters.

First, anthropoid lenses contain a series of tryptophan-derived compounds (the most abundant being 3-hydroxy-L-kynurenine glucoside) that screen out most light below about 400 nm in wavelength (i.e., UV light; Bova et al., 1999; Van Heyningen, 1973). For example, anthropoid lenses typically transmit less than 0.0001% of incident 360 nm UV light (Cooper and Robson, 1969b). This condition differs sharply from that seen in most other mammalian lenses (e.g., including those of many rodents, artiodactyls, perissodactyls, and carnivorans), which are relatively transparent to UV light above 320 nm in wavelength (Cooper and Robson, 1969a; Gorgels and van Norren, 1992). Indeed, many rodents with UV-transmitting lenses (e.g., gerbils [*Meriones unguiculatus*], hamsters [*Mesocricetus auratus*, *Phodopus sungorus*], mice [*Mus musculus*], and gophers [*Thomomys bottae*]) possess short-wavelength photopigments that absorb maximally within the UV range (Calderone and Jacobs, 1999; Jacobs, 1992; Jacobs and Deegan, 1994). The only mammals known to exhibit UV-opaque lenses similar to those of anthropoids are diurnal sciurids (Cooper and Robson, 1969a). Like diurnal anthropoids, the lenses of diurnal squirrels also contain tryptophan-derived UV screening compounds (in the case of squirrels, primarily N-acetyl-3-hydroxy-L-kynurenine; Van Heyningen, 1971). Because anthropoid photoreceptor pigments readily absorb UV light through

off-peak or beta-band (secondary peak) absorption (Jacobs, 1992; Stark and Tan, 1982), near-UV light (between about 400 nm and 320 nm) is “visible” to anthropoid photoreceptors in the absence of lenticular screening. This phenomenon accounts for the observation that humans who have had their lenses removed (i.e., “aphakics”) readily perceive near-UV light in behavioral tests (Stark, 1987; Stark et al., 1994). Accordingly, it appears that anthropoids have sacrificed the ability to perceive a large portion of the spectrum that is visible to most other mammals for the sake of improving visual acuity.

While the anthropoid lens screens out most UV light, the retinas of diurnal anthropoids play an important role in selectively filtering out blue and violet “visible” light. In all anthropoids except *Aotus*, the central visual area (i.e., the central region of the retina which supports the highest visual acuity and the best ability to discriminate between colors) contains high concentrations of the carotenoids lutein and zeaxanthin (Bone et al., 1985; Nussbaum et al., 1981; Polyak, 1957; Rodieck, 1988). Both compounds absorb light maximally in the blue region of the visible spectrum (about 460 nm), but transmit nearly all light of wavelengths greater than 550 nm (Nussbaum et al., 1981; Stark, 1987). As a result, these carotenoid accumulations are visible upon gross examination as a “yellow spot,” or macula lutea. In addition to diurnal anthropoids, a macula lutea is also visible in freshly fixed retinas of *Tarsius spectrum* (Castenholz, 1984; Hendrickson et al., 2000; Polyak, 1957), suggesting that this feature was present in the last common ancestor of living haplorhines.

Macular pigments are primarily concentrated in the fiber of Henle layer and the inner and outer plexiform layers of the fovea and the perifoveal retina (Snodderly et al., 1984a, 1984b). As a result, all light passing to the photoreceptor outer segments in the fovea must first pass through the yellow macular pigments. By selectively filtering out blue light, these pigments reduce chromatic aberration and thus improve visual acuity in the central part of the visual field (Douglas and Marshall, 1999; Walls, 1942). In this capacity, the macula lutea functionally resembles the yellow corneas of some fishes, the yellow lenses of diurnal tree shrews and sciurids, and the diffuse yellow retinal pigments of some lizards (Barbour et al., 2002; Douglas and Marshall, 1999; Walls, 1942). Additional evidence suggests that macular pigments also protect foveal photoreceptors from oxidative damage caused by short-wavelength visible light (Thomson et al., 2002; Wooten and Hammond, 2002). The macula lutea may thus be important for maintaining the long-term functionality of the fovea. Significantly, both of the functional roles proposed for the macula lutea

(i.e., reduction of chromatic aberration and protection of the fovea) are relevant only in the context of vision under diurnal conditions (see section on Implications for Anthropoid Origins).

Central Retinal Anatomy

The central retinas of most mammals exhibit varying degrees of specialization for increased visual acuity relative to the retinal periphery. For example, the central retinas of all primates are distinguished by a lack of large blood vessels running on their inner surface (Johnson, 1901; Polyak, 1957; Walls, 1942; Wolin and Massopust, 1970). This largely avascular region of the central retina in primates has been termed the “central visual area” by Wolin and Massopust (1970). The exclusion of larger vessels from the central visual area enhances visual acuity because blood can absorb and scatter light before it reaches the retina’s photoreceptor layer (Adams and Horton, 2002; Weale, 1966). All primates also show a regular increase in both the density of photoreceptors and ganglion cells, and the lengths of photoreceptor outer segments from the retinal periphery to the central visual area (Allman, 1977; Andrade da Costa and Hokoç, 2000; Curcio et al., 1990; DeBruyn et al., 1980; Dkhissi-Benyahya et al., 2001; Hendrickson et al., 2000; Rohen and Castenholz, 1967; Stone and Johnston, 1981; Troilo et al., 1993; Wikler and Rakic, 1990; Wilder et al., 1996; Yamada et al., 1998, 2001). By contrast, the sizes of ganglion cell dendritic arbors and receptive field centers decrease steadily from the periphery of the retina to its visual center (Ghosh et al., 1996; Hubel and Wiesel, 1960; Goodchild et al., 1996). Increased photoreceptor and ganglion cell densities and decreased receptive field center diameters are functionally interrelated. Each of these features helps to confer greater spatial acuity in the central visual area by increasing the number of independent sampling units per degree of visual angle that can transmit information to the brain (Berkley, 1976). Similarly, increased photoreceptor outer segment length is probably an adaptation for increasing visual acuity by reducing the effects of chromatic aberration (Walls, 1942).

Despite these general similarities in retinal organization, primates vary considerably in the degree to which retinal centralization is developed. Some primates (e.g., *Cheirogaleus* and *Microcebus*) show no specialization of the central visual area beyond that described above (Rohen and Castenholz, 1967). Most strepsirrhines however (e.g., *Propithecus*, *Indri*, *Varecia*, *Lemur*, *Otolemur*, and *Galago*) exhibit a dome-shaped thickening of the central visual area due to a

greatly increased number of photoreceptors, ganglion cells, and supporting cells (DeBruyn et al., 1980; Rohen and Castenholz, 1967). Such a thickening marks the presence of an “area centralis” (Polyak, 1957), defined by Walls as a “circumscribed retinal area in which the retina is so constructed as to afford a marked local increase in resolving power” (Walls, 1942, p. 181).

By contrast, in all haplorhines except *Aotus*, the central visual area is occupied by a circular or ovate pit called the “fovea.” The consistent presence of retinal foveae is unique to tarsiers and anthropoids among mammals (but see DeBruyn et al., 1980 and Stone and Johnston, 1981 for reports of rare foveal development in galagos), although foveae are commonly present in the retinas of many non-mammalian vertebrates (Fite and Rosenfield-Wessels, 1975; Inzunza et al., 1991; Makaretz and Levine, 1980; Ross, this volume; Walls, 1942). The haplorhine fovea is formed perinatally and is typically the last region of the retina to reach maturity (Hendrickson and Kupfer, 1976; Provis et al., 1998). Foveal development in diurnal anthropoids begins when a large number of ganglion cells accumulate over a region of the central retina that is devoid of rods. This increased number of ganglion cells produces a localized thickening of the retina that (except for the lack of any rods) closely resembles a strepsirrhine area centralis. When this foveal anlage reaches a critical thickness, the cells of the inner and outer nuclear layers begin to migrate away from the center of the fovea while cones move inwardly and become more densely packed (Hendrickson and Kupfer, 1976; Leventhal and Schall, 1989; Leventhal et al., 1989). As cells of the inner and outer nuclear layers are displaced, they maintain their initial synaptic relationships and carry their blood supply away from the foveal center. The center of the haplorhine fovea is thus typically free of all internal vasculature, including capillaries (Ferraz de Oliveira and Ripps, 1968; Polyak, 1957; Provis et al., 1998; Wolin and Massopust, 1970), and most of the neuroretinal cells that receive their blood supply from the central artery of the retina rather than choroidal vessels (i.e., all but the photoreceptors) are absent from the foveal center (Leventhal et al., 1989; Silveira et al., 1989; Wilder et al., 1996).

In diurnal anthropoids, the adult fovea is entirely free of rods (Andrade da Costa and Hokoç, 2000; Polyak, 1957; Walls, 1942; Wikler and Rakic, 1990) and the avascular foveal center is composed of a densely-packed hexagonal lattice of cones (Hirsch and Hylton, 1984). Maximum foveal cone densities in diurnal anthropoids are very high regardless of eye size, typically ranging between about 100,000–400,000 cones mm² (Table 3). These high foveal

Table 3. Peak retinal cone densities in various mammalian taxa

Scientific name	Common name	Activity pattern	Maximum cone density (cones/mm ²)	Source
<i>Alouatta caraya</i>	Black howler monkey	Diurnal	359,000–430,000	Franco et al, 2000
<i>Callithrix jacchus</i>	Common marmoset	Diurnal	152,000–360,000	Troilo et al., 1993; Wilder et al., 1996; Martin and Grünert, 1999; Franco et al, 2000
<i>Homo sapiens</i>	Human	Diurnal	100,000–324,000	Curcio et al., 1990
<i>Macaca nemestrina</i>	Pig-tailed macaque	Diurnal	180,000–261,000	Packer et al., 1989
<i>Cercopithecus aethiops</i>	Vervet monkey	Diurnal	179,000–210,000	Perry and Cowey, 1985
<i>Cebus apella</i>	Tufted capuchin	Diurnal	141,000–176,000	Andrade da Costa and Hokoç, 2000; Franco et al., 2000
<i>Macaca mulatta</i>	Rhesus macaque	Diurnal	134,000–160,000	Perry and Cowey, 1985; Wikler and Rakic, 1990
<i>Saimiri ustius</i>	Squirrel monkey	Diurnal	90,000–140,000	Franco et al., 2000
<i>Saguinus midas</i>	Golden-handed tamarin	Diurnal	129,000–135,000	Franco et al., 2000
<i>Macaca fascicularis</i>	Crab-eating macaque	Diurnal	95,000–104,000	Perry and Cowey, 1985
<i>Tarsius spectrum</i>	Spectral tarsier	Nocturnal	~50,000	Hendrickson et al., 2000
<i>Meriones unguiculatus</i>	Mongolian gerbil	Diurnal	45,000–50,000	Govardovskii et al., 1992
<i>Spermophilus beecheyi</i>	California ground squirrel	Diurnal	41,800–50,000	Long and Fisher, 1983; Kryger et al., 1998
<i>Tupaia glis</i>	Tree shrew	Diurnal	36,000	Müller and Peichl, 1989
<i>Felis catus</i>	Domestic cat	Cathemeral	~27,000	Steinberg et al., 1973; Linberg et al., 1998
<i>Tachyglossus aculeatus</i>	Echidna	Cathemeral	22,000	Young and Pettigrew, 1991
<i>Macropus eugenii</i>	Tammar wallaby	Cathemeral	~18,700	Hemmi and Grünert, 1999

<i>Mus musculus</i>	House mouse	Cathemeral	11,000–18,000	Szél et al., 1992, 1994
<i>Oryctolagus cuniculus</i>	Old World rabbit	Cathemeral	~14,000	Juliussen et al., 1994
<i>Phodopus sungorus</i>	Siberian dwarf hamster	Cathemeral	10,000–13,000	Calderone and Jacobs, 1999; Lukáts et al., 2002
<i>Phoca hispida</i>	Ringed seal	Cathemeral	10,000–11,000	Peichl and Moutairou, 1998
<i>Mesocricetus auratus</i>	Golden hamster	Nocturnal	~10,000	Calderone and Jacobs, 1999
<i>Apodemus sylvaticus</i>	Wood mouse	Nocturnal	~9,500	Szél et al., 1994
<i>Otolemur garnettii</i>	Garnett's bush baby	Nocturnal	7,500–8,500	Wikler and Rakic, 1990
<i>Didelphis virginiana</i>	North American opossum	Nocturnal	8,000	Kolb and Wang, 1985
<i>Microcebus murinus</i>	Gray mouse lemur	Nocturnal	7,500–8,000	Dkhissi-Benyahya et al., 2001
<i>Mus spicilegus</i>	Mound builder mouse	Cathemeral	~7,000	Szél et al., 1994
<i>Aotus trivirgatus</i>	Northern owl monkey	Nocturnal	5,000–6,000	Wikler and Rakic, 1990
<i>Apodemus microps</i>	Herb field mouse	Nocturnal	~5,000	Szél et al., 1994
<i>Saccostomus campestris</i>	Pouched mouse	Nocturnal	~4,500	Lukáts et al., 2002
<i>Cricetomys</i> spp.	African giant pouched rat	Nocturnal	3,000–4,300	Peichl and Moutairou, 1998
<i>Didelphis marsupialis</i>	South American opossum	Nocturnal	2,400–2,600	Ahnelt et al., 1995

Notes: The peak cone density is expressed as the maximum number of cones per square millimeter. Density ranges for each species represent the full range of intraspecific variation reported by the author(s). Activity pattern source and definitions are the same as for Table 1.

cone densities in diurnal anthropoids contrast with lower maximum cone densities in nocturnal strepsirrhines (range: 5,000–8,500 cones mm²), tarsiers (50,000 cones mm²), and nonprimate mammals (range: 2,400–50,000 cones mm²) (Table 3). Unfortunately, maximum cone densities of diurnal anthropoids and day-active (i.e., diurnal and cathemeral) strepsirrhines cannot be directly compared because there is little published quantitative data on the retinal morphology of lemuriform primates. However, qualitative assessments of retinal morphology (Kolmer, 1930; Ordy and Samorajski, 1968; Wolin and Massopust, 1970; Woollard, 1927) and studies of retinal electrophysiology (Alfieri et al., 1976) in *Lemur*, *Eulemur*, *Varecia*, *Propithecus*, and *Indri* agree that central rod:cone ratios in these genera are relatively high. These observations thus suggest that central cone densities in diurnal and cathemeral lemuriforms are much lower than in diurnal anthropoids.

As is evident from its lower peak cone densities, *Tarsius* does not exhibit the pattern of foveal organization found in diurnal members of the Haplorhini. In tarsiers, the fovea contains rods as well as cones, and some capillaries may be found within the foveal slope (Hendrickson et al., 2000; Polyak, 1957; Ross, this volume; Wolin and Massopust, 1970). Even though tarsiers have higher peak cone densities than most other mammals (Table 3), peak ganglion cell densities of tarsiers are comparable to those of other nocturnal primates and are significantly lower than those of diurnal anthropoids (Tetreault et al., this volume). This finding is supported by published ratios of the number of cells in the outer nuclear layer (photoreceptors) and ganglion cell layer (ganglion cells and displaced amacrine cells) of the haplorhine perifoveal retina, which range from 7:1 to 8:1 in tarsiers, and from 1:1 to 1:2 in diurnal anthropoids (Castenholz, 1984; Rohen and Castenholz, 1967). Thus, although a tarsier's fovea structurally resembles that of a diurnal anthropoid, its large proportion of rods and smaller proportion of surrounding ganglion cells make it functionally more similar to the area centralis of a nocturnal strepsirrhine (e.g., *Galago senegalensis*; Ross, this volume; Wolin and Massopust, 1970).

In contrast to all other haplorhines, *Aotus* only rarely exhibits a retinal fovea (Lima et al., 1996; Ogden, 1994; Walls, 1953; Webb and Kaas, 1976). When present (about 1 in 10 individuals; Ogden, 1994), owl monkey foveae are relatively small and shallow. These "rudimentary" foveae differ greatly from the foveae of diurnal anthropoids in containing a large proportion of rods and admitting some retinal capillaries to supply a continuous layer of ganglion cells across the fovea (Polyak, 1957; Webb and Kaas, 1976). However, most

individuals of *Aotus* exhibit an area centralis that is reminiscent of those seen in diurnal strepsirrhines and marked by a localized thickening of the retina (Ferraz de Oliveira and Ripps, 1968; Lima et al., 1996; Murray et al., 1973; Ogden, 1975; Webb and Kaas, 1976). The variable occurrence of a fovea in *Aotus* and the difficulty of detecting a foveal depression without the use of histological sections (Prince, 1956) may explain the considerable historical disagreement regarding the existence of a fovea in this genus (e.g., compare Polyak, 1957 and Jones, 1965).

Because *Aotus* exhibits relatively low peak cone densities, it seems unlikely that the rudimentary foveae seen in some owl monkeys could significantly augment visual acuity. Similarly, while *Tarsius* demonstrates higher peak cone densities than other nocturnal primates and most nonprimate mammals (Hendrickson et al., 2000; Table 3), the relatively low central ganglion cell densities (Tetreault et al., this volume) and low ratio of ganglion cells to photoreceptors in the central retina (Rohen and Castenholz, 1967) suggest that tarsier foveae probably do not accommodate high visual acuity comparable to that of diurnal anthropoids. This conclusion is further supported by calculations of the gain produced by foveal spreading of the retinal image in *Tarsius* (cf. Walls, 1942), which indicate that potential gains in acuity through magnification of the foveal image would be trivial (Ross, this volume). By contrast, the all-cone foveae of diurnal anthropoids provide the primary anatomical basis for their extremely high visual acuity (Tables 1 and 2). Indeed, the centrifugal displacement of most non-photoreceptor retinal cells during foveal formation is probably itself an adaptation for increasing central visual acuity. Although cells located on the retina's inner surface (i.e., "in front" of the photoreceptors) are transparent and relatively homogenous, they may be responsible for minute variations in refractive index that slightly degrade retinal resolution. Any such variation would be eliminated over the fovea with the relocation of these cell layers to perifoveal regions that are less specialized for high visual acuity (Rodieck, 1988). Furthermore, the centrifugal displacement of non-photoreceptor neuroretinal cells during foveal development ensures that blood in the capillaries supplying these cells does not scatter light reaching the fovea (Weale, 1966; Wolin and Massopust, 1970). The exclusion of most inner retinal cells as well as capillaries from the foveal center thus affords incident light rays an unobstructed path to the photoreceptor layer.

While the formation of the foveal pit probably plays an important role in eliminating minute refractive errors within the retina, it is the very high density

and specialized connectivity of foveal cones that is most critical for producing the high visual acuity characteristic of diurnal anthropoids. Not only are cones present in unusually high densities per unit of visual angle within the anthropoid fovea, but each foveal cone stimulates two midget ganglion cells (i.e., one ON and one OFF midget ganglion cell; Schein, 1988; Wässle and Boycott, 1991; Wässle et al., 1989, 1990, 1994; Wilder et al., 1996). Therefore, every foveal cone has two “private lines” to the brain because midget ganglion cells are connected via a single cone bipolar cell to only one cone. Each foveal cone thus represents an independent sampling unit of the retinal image, and the densely-packed foveal cone mosaic is able to pass highly detailed spatial information to the brain. In fact, all other specializations of the diurnal anthropoid eye for high visual acuity (e.g., large eye size, short-wavelength screening, etc.) may be viewed as means of ensuring that the foveal photoreceptor mosaic is able to sample a relatively large and clear retinal image. By contrast, even the best developed central visual areas of strepsirrhines and nocturnal haplorhines are incapable of supporting such high degrees of visual acuity because ganglion cells in the central retina have receptive field centers composed of *multiple* photoreceptors (Dkhissi-Benyahya et al., 2001; Rohen and Castenholz, 1967; Yamada et al., 1998). This inward convergence of more than one photoreceptor cell onto a single ganglion cell via bipolar cell (or AII amacrine) intermediaries is referred to as “retinal summation” (Kay and Kirk, 2000; Walls, 1942). As a rule, foveal cones of diurnal anthropoids exhibit no summation in the midget (parvocellular) system, while the central photoreceptors of other primates summate more extensively (Goodchild et al., 1996; Polyak, 1957; Rohen and Castenholz, 1967; Schein, 1988; Walls, 1942).

OSTEOLOGICAL CORRELATES OF RETINAL SUMMATION AND VISUAL ACUITY IN EXTANT PRIMATES

Orbit Size

As is evident from the discussion above, the majority of the adaptations shared by anthropoid primates for increased visual acuity are found in the soft tissues of the eye, and thus leave no trace in the fossil record. Questions about the timing and ecological context of the appearance of most of these soft-tissue traits (e.g., the macula lutea) can only be addressed through parsimony-based analyses of the distribution of relevant character states among extant species of primates.

However, because of the conformation between the eye and the bony orbit in which it is enclosed (Schultz, 1940; Figure 1), orbit size in fossil taxa with sufficiently-preserved crania may be used to estimate the absolute size of the eye. This correlation between orbit size and eye size has been exploited for a number of years in analyses of fossil primate taxa that seek to determine activity pattern based on relative orbit size (Heesy and Ross, 2001; Kay and Cartmill, 1977; Kay and Kirk, 2000; Walker, 1967). Such studies have found that orbit size in Eocene to Miocene anthropoid species (e.g., *Aegyptopithecus zeuxis*, *Apidium phiomense*, *Catopithecus browni*, *Dolichocebus gaimanensis*, *Simonsius grangeri*, *Proteopithecus sylviae*, and *Tremacebus harringtoni*) fall within the range of extant anthropoids of similar cranial length (Heesy and Ross, 2001; Kay and Kirk, 2000; Simons, 1995, 1997, 2001; Simons and Rasmussen, 1996). Accordingly,

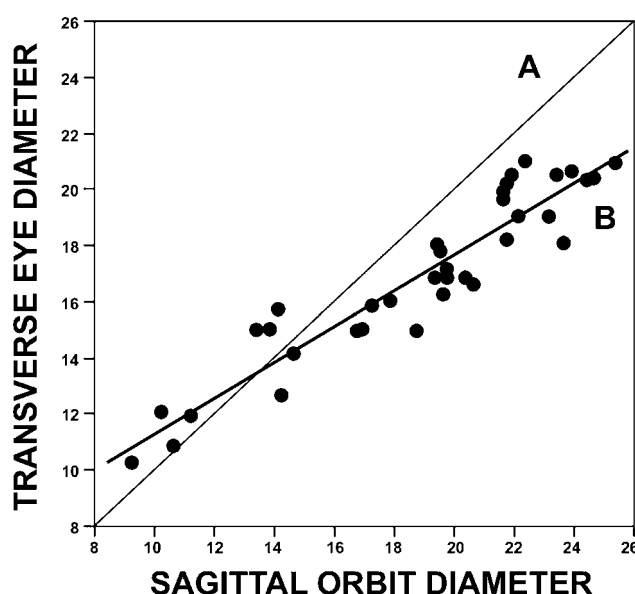


Figure 1. Bivariate plot of the transverse eye diameter (y -axis) versus maximum orbit diameter as measured in a parasagittal plane (x -axis). Both values are in mm. Line A represents equal orbit diameter and eye diameter. Line B is a least-squares regression fit to the data ($Y = 4.90 + 0.64 X$; $r^2 = 0.88$). Each data point represents a single individual. Measurements were taken from formalin-preserved specimens ($n = 36$) in the comparative anatomy collections of Duke University and SUNY Stony Brook. Represented genera include: *Alouatta*, *Aotus*, *Ateles*, *Cebus*, *Eulemur*, *Galago*, *Hylobates*, *Lagothrix*, *Lemur*, *Leontopithecus*, *Loris*, *Macaca*, *Nycticebus*, *Otolemur*, *Papio*, *Perodicticus*, *Saguinus*, *Saimiri*, and *Varecia*.

all known fossil anthropoids appear to have resembled extant primates in having relatively large eyes when compared to many nonprimate mammals (Hughes, 1977; Kay and Cartmill, 1977; Ritland, 1982; Stephan et al., 1984; Walls, 1942). Although these data on relative orbit size do not reveal a great deal regarding the visual acuity of fossil anthropoids, they do indicate that as early as the late Eocene, eye size in anthropoids was large enough to have *potentially* allowed the very high visual acuity that characterizes extant diurnal anthropoids.

Optic Foramen Size

In addition to the close relationship between the size of the eye and the size of the orbit (Figure 1), there is also a close relationship between the cross-sectional areas of the optic foramen and the optic nerve (Figure 2). Because optic nerve cross-sectional area is highly correlated with the total number of axons in the optic nerve in both intraspecific (Figure 3; Arey et al., 1942; Jonas et al., 1992; Linke and Roth, 1990; Quigley et al., 1991) and interspecific (Figure 4) comparisons, the area of the optic foramen may be used as an index of the total number of axons within the optic nerve. The number of centrifugal fibers

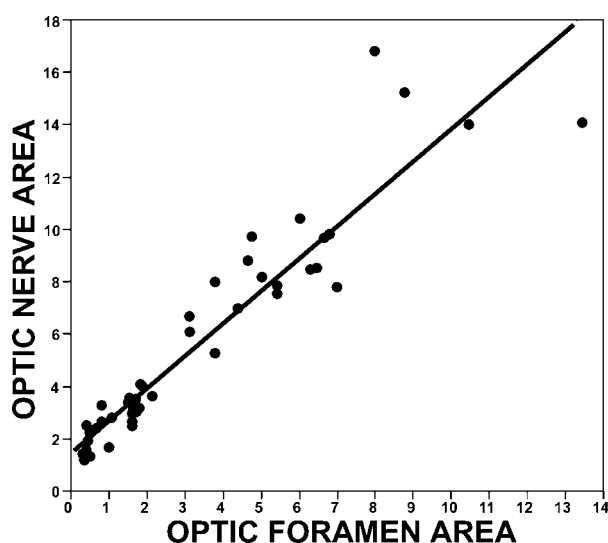


Figure 2. Bivariate plot of the cross-sectional area of the optic nerve (y -axis) versus the cross-sectional area of the optic foramen (x -axis) in 45 primate species. Both values are in mm^2 . Optic foramen areas are taken from Table 4. Nerve cross-sectional areas are taken from Table 1 of Stephan et al. (1984). Least-squares regression: $Y = 1.47 + 1.24X$; $r^2 = 0.88$. Spearman Rho = 0.95 ($p < 0.0001$).

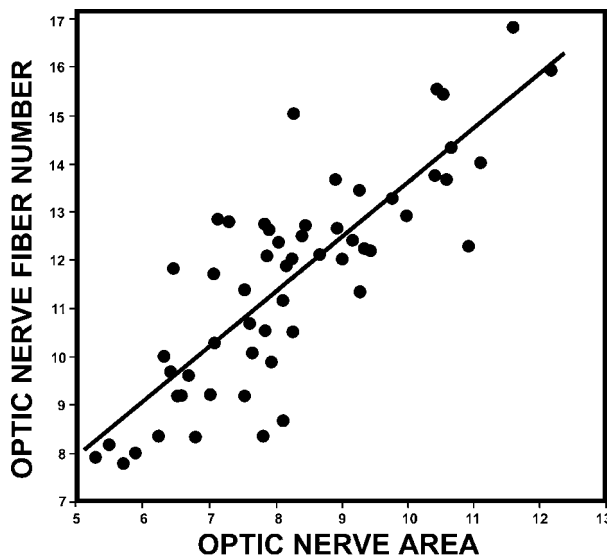


Figure 3. Bivariate plot of the number of optic nerve fibers (=retinal ganglion cell axons, in tens of thousands; y -axis) versus the cross-sectional area of the retrobulbar optic nerve (in mm^2 ; x -axis) for 56 normal eyes from 56 human subjects. Data are redrawn from Figure 4 of Jonas et al. (1992), with permission. Least-squares regression: $Y = 218,000 + 114,000 X$; $r^2 = 0.67$.

(i.e., efferent, from brain to retina) in the optic nerves of most vertebrate species is estimated to be a tiny fraction of the total number of nerve fibers (i.e., usually less than 1%–2%; Bennis et al., 2001; Binggeli and Paule, 1969; Brooke et al., 1965; Cowan and Powell, 1963; Freeman and Watson, 1978; Kirby et al., 1982). Accordingly, these data indicate that the cross-sectional areas of both the optic nerve and optic foramen are proportional to the total number of ganglion cells in the retina. All afferent visual information is transmitted from the retina to the brain via the long axons of retinal ganglion cells. It is these retinal ganglion cell axons and their supporting neuroglial elements that comprise the bulk of the optic nerve.

The Optic Foramen Quotient (OFQ)

As noted previously, visual acuity is a function of both the size and quality of the retinal image, as well as the spatial distribution of the individual sampling units of the retina (i.e., ganglion cell receptive field centers) upon which the retinal image is projected. Retinal image size is determined by the size of the eye and

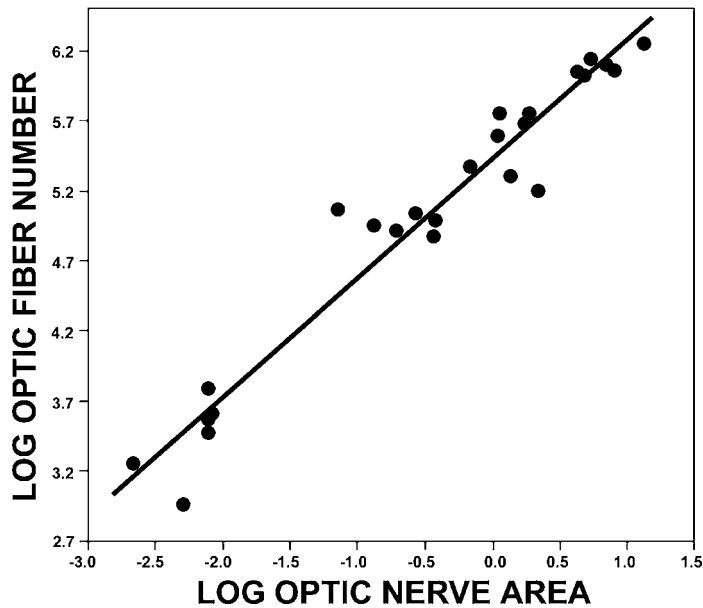


Figure 4. Bivariate plot of the number of optic nerve fibers (y -axis) versus the cross-sectional area of the optic nerve (mm^2 ; x -axis) in 25 mammal species. All values have been natural-log transformed. Represented taxa include: *Homo sapiens* (Jonas et al., 1992), *Macaca fascicularis* (Sanchez et al., 1986), *Macaca mulatta* (Morrison et al., 1990), *Papio anubis* (Fischer and Kirby, 1991; Stephan et al., 1984), *Cercopithecus aethiops* (Herbin et al., 1997; Stephan et al., 1984), *Cebus apella* and *Aotus* sp. (Lima et al., 1996; Stephan et al., 1984), *Tupaia belangeri* (Drenhaus et al., 1997), *Felis catus* (Williams et al., 1986), *Mesocricetus auratus* (Tiao and Blakemore, 1976), *Rattus norvegicus* (Fukuda et al., 1982), *Tamias sibiricus* (Wakakuwa et al., 1987), *Oryctolagus cuniculus* (Vaney and Hughes, 1976), *Sminthopsis crassicaudata* (Arrese et al., 1999), *Trichosurus vulpecula* (Freeman and Watson, 1978), *Didelphis marsupialis* (Hokoç and Oswaldo-Cruz, 1978), *Didelphis virginiana* (Kirby et al., 1982), *Setonix brachyurus* (Brackevelt et al., 1986), *Tarsipes rostratus* (Dunlop et al., 1994), *Cryptomys hottentotus*, *Spalax ehrenbergi*, *Spalax leucodon*, *Talpa europaea*, *Crocidura suaveolens*, and *Sorex araneus* (Herbin et al., 1995). Least-squares regression: $Y = 5.43 + 0.85 X$; $r^2 = 0.95$. Spearman Rho = 0.95 ($p < 0.0001$).

the configuration of the cornea and lens, while the sampling density of the retina is determined primarily by photoreceptor density and retinal summation (Berkley, 1976; Polyak, 1957; Walls, 1942). Because optic foramen cross-sectional area is proportional to the number of retinal ganglion cell axons in the optic nerve, the size of the optic foramen provides a relative means of

estimating the total number of individual sampling units in the retina. However, optic foramen area in itself is not useful for drawing conclusions regarding visual acuity because the total number of retinal sampling units can change simply as a function of eye size. In other words, larger eyes *ipso facto* have larger retinas, which may or may not have proportionally greater numbers of ganglion cells depending on the sampling density. Among primates, optic foramen area is positively correlated with orbit area (Figure 5), but the size of the area centralis or fovea is very similar across taxa (Franco et al., 2000; Rohen and Castenholz, 1967). These data suggest that as eye size increases in primates, additional ganglion cells must be added to sample the proportionally larger retinal periphery. As a result, if one makes interspecific comparisons of optic foramen size independent of an estimator of eye size, one does not know whether species differences are the result of changes in “fineness of grain” with which the retinal image is sampled by the photoreceptor mosaic in the central retina (and hence, changes in visual acuity) or whether such differences are the result of changes in eye size alone (not *necessarily* associated with changes in acuity). However, if optic foramen size is adjusted for eye size, then the

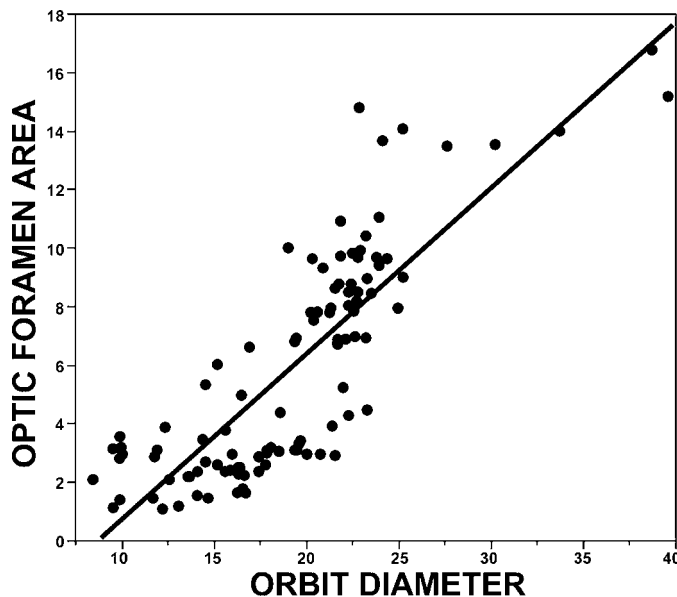


Figure 5. Bivariate plot of optic foramen area (mm^2 ; y-axis) versus sagittal orbit diameter (mm; x-axis) in 103 primate species. Least-squares regression: $Y = -4.95 + 0.57 X$; $r^2 = 0.68$. Spearman Rho = 0.83 ($p < 0.0001$).

resulting values may be viewed as an estimate of the average density of sampling units within the retina. While this methodology cannot account for variation in visual acuity caused by differences in the size of the retinal image or different patterns of photoreceptor packing across the retina, it may serve as a rough index of relative visual acuity in taxa with similar ocular and retinal bauplans (e.g., within primates; see below).

Accordingly, as a first step in making interspecific comparisons of relative optic foramen size, we have calculated an Optic Foramen Index, or OFI (Kay and Kirk, 2000). The OFI is calculated using the following formula:

$$\text{OFI} = (\text{Optic Foramen Area} / \text{Orbit Area}) \times 100$$

Optic foramen areas, orbit areas, and OFIs for 93 species of extant primates are presented in Table 4.

While the OFI is a useful first approximation of optic foramen size relative to eye size, it is necessary to adjust the OFI for body size because orbit size progressively overestimates eye size as body size increases (Figure 1; Kay and Kirk, 2000; Schultz, 1940). This “looser fit” between the eye and the orbit at larger body sizes is responsible for observed decreases in OFI with increasing prosthion–inion (P–I) length (Figure 6). To adjust the OFI for this size-related error, we have fit separate natural logarithm-transformed least-squares regression lines to the subgroups of extant nocturnal strepsirrhines and diurnal haplorhines in a bivariate plot of OFI versus P–I length (Figure 6). These two groups of species have relatively little intragroup variation in retinal anatomy, eye morphology, and visual capabilities (discussed previously). By fitting regression lines to groups with similar visual bauplans, these lines represent our best attempt at creating a line of “visual functional equivalence” for the purpose of making interspecific comparisons across a wide range of body sizes. Deviations of species means (extant taxa) or individual specimens (fossil taxa) from the regression lines were expressed as Optic Foramen Quotients (OFQs) using the following formula:

$$\text{OFQ} = [(\text{Observed OFI} - \text{Expected OFI}) / \text{Expected OFI}] \times 100$$

The OFQ is thus the observed OFI expressed as a percentage of the OFI “expected” for a diurnal anthropoid or nocturnal strepsirrhine (depending on which empirical regression formula is used) of similar P–I length. OFQs based on the diurnal anthropoid regression (“daOFQs”) and the nocturnal strepsirrhine regression (“nsOFQs”) are presented in Table 5.

Table 4. Optic foramen indices (OFIs) of extant species

Species	N	Activity	P-I length	SD	OF area	SD	Orbit area	SD	OFI	SD
<i>Allenopithecus nigroviridis</i>	2	Diurnal	98.5	1.7	7.0	0.1	298.7	4.3	2.33	0.08
<i>Alouatta palliata</i>	6	Diurnal	109.8	8.8	5.3	0.5	382.3	26.8	1.38	0.12
<i>Alouatta seniculus</i>	6	Diurnal	108.0	7.2	8.0	1.7	359.4	21.0	2.21	0.37
<i>Aotus trivirgatus</i>	10	Nocturnal	60.5	1.3	3.1	0.4	297.4	17.8	1.05	0.16
<i>Arctocebus aureus</i>	1	Nocturnal	49.5		1.2		134.8		0.89	
<i>Arctocebus calabarensis</i>	7	Nocturnal	53.8	1.8	1.6	0.8	156.6	11.7	1.01	0.52
<i>Ateles fusciceps</i>	7	Diurnal	115.0	4.9	9.7	1.1	469.4	52.4	2.08	0.30
<i>Ateles geoffroyi</i>	7	Diurnal	107.6	3.6	9.7	1.2	409.4	12.7	2.38	0.32
<i>Ateles paniscus</i>	10	Diurnal	107.7	3.6	6.9	1.2	385.9	46.8	1.79	0.24
<i>Avahi laniger</i>	10	Nocturnal	53.4	1.0	1.7	0.2	219.3	16.4	0.77	0.10
<i>Cacajao melanocephalus</i>	6	Diurnal	91.8	2.3	6.9	0.8	372.1	44.8	1.85	0.21
<i>Callicebus moloch</i>	10	Diurnal	59.7	2.5	3.5	0.6	163.9	17.7	2.14	0.25
<i>Callicebus torquatus</i>	6	Diurnal	65.8	1.5	3.8	0.4	192.0	7.1	1.99	0.17
<i>Callimico goeldii</i>	7	Diurnal	52.3	1.6	3.9	0.5	121.2	10.6	3.23	0.39
<i>Callithrix argentata</i>	6	Diurnal	45.2	0.6	2.8	0.5	77.3	2.5	3.68	0.66
<i>Callithrix jacchus</i>	9	Diurnal	43.7	1.2	3.2	0.3	72.4	6.1	4.39	0.61
<i>Cebuella pygmaea</i>	10	Diurnal	34.1	1.1	2.1	0.3	56.3	2.1	3.75	0.50
<i>Cebus albifrons</i>	6	Diurnal	87.5	3.5	7.5	0.7	328.7	34.1	2.31	0.24
<i>Cebus capucinus</i>	7	Diurnal	95.1	1.9	7.8	0.5	337.0	47.7	2.36	0.39
<i>Cercopithecus cephus</i>	8	Diurnal	95.9	5.5	9.3	1.5	344.0	29.7	2.73	0.46
<i>Cercopithecus mitis</i>	6	Diurnal	108.9	9.7	8.5	0.8	409.9	56.4	2.11	0.35
<i>Cercopithecus petaurista</i>	7	Diurnal	92.7	5.7	7.8	1.2	324.4	22.5	2.43	0.39
<i>Cheirogaleus major</i>	9	Nocturnal	55.3	1.6	1.5	0.2	169.8	16.6	0.87	0.14
<i>Cheirogaleus medius</i>	2	Nocturnal	43.3	0.5	1.1	0.03	117.9	1.4	0.94	0.04
<i>Chiropotes satanas</i>	6	Diurnal	83.4	4.1	6.8	1.0	295.3	16.0	2.32	0.34
<i>Chlorocebus aethiops</i>	8	Diurnal	102.9	7.4	7.8	0.8	327.8	16.6	2.39	0.24
<i>Colobus guereza</i>	6	Diurnal	111.6	5.1	9.8	1.0	376.8	35.5	2.60	0.34
<i>Colobus polykomos</i>	6	Diurnal	111.1	5.5	8.5	1.2	391.2	47.0	2.20	0.34

Table 4. Continued

Species	N	Activity	P-I length	SD	OF area	SD	Orbit area	SD	OFI	SD
<i>Daubentonia madagascariensis</i>	6	Nocturnal	86.7	3.2	2.9	0.2	364.9	17.3	0.81	0.07
<i>Erythrocebus patas</i>	6	Diurnal	125.7	13.9	9.8	1.3	400.6	50.4	2.49	0.44
<i>Eulemur fulvus</i>	18	Cathemeral	90.4	3.2	3.4	0.6	306.4	34.8	1.14	0.25
<i>Eulemur macaco</i>	3	Cathemeral	89.5	2.7	3.1	0.3	295.8	18.4	1.06	0.12
<i>Eulemur mongoz</i>	4	Cathemeral	79	1.5	3.0	0.2	251.7	10.8	1.21	0.13
<i>Eulemur rubriventer</i>	9	Cathemeral	86.2	2.2	3.0	0.2	317.7	24.8	0.96	0.13
<i>Euoticus elegantulus</i>	7	Nocturnal	48.6	1.1	2.9	0.5	240.0	16.4	1.21	0.19
<i>Galago alleni</i>	6	Nocturnal	49.6	1.4	2.3	0.3	211.0	12.2	1.08	0.12
<i>Galago matschiei</i>	1	Nocturnal	45.2		3.0		201.1		1.49	
<i>Galago moholi</i>	11	Nocturnal	38.3	2.2	2.2	0.3	147.9	9.4	1.49	0.19
<i>Galago senegalensis</i>	6	Nocturnal	43.9	2.5	2.6	0.6	183.9	32.2	1.45	0.34
<i>Galagoides demidoff</i>	15	Nocturnal	37.3	1.7	1.5	0.3	109.2	11.7	1.36	0.27
<i>Galagoides zanzibaricus</i>	10	Nocturnal	40.8	1.8	2.1	0.3	125.8	16.6	1.70	0.21
<i>Gorilla gorilla</i>	12	Diurnal	268.1	37.0	15.0	2.2	1236.0	93.2	1.23	0.20
<i>Hapalemur griseus</i>	21	Diurnal	65.5	2.5	2.4	0.4	200.4	17.8	1.21	0.14
<i>Hapalemur simus</i>	2	Cathemeral	81.9	0.4	3.0	0.1	340.4	41.6	0.88	0.13
<i>Homo sapiens</i>	1	Diurnal	186.0		14.0		721.1		1.88	
<i>Hylobates klossii</i>	8	Diurnal	96.4	2.5	7.8	0.8	355.3	20.5	2.20	0.19
<i>Hylobates lar</i>	10	Diurnal	103.8	3.5	8.5	0.8	434.8	37.0	1.95	0.22
<i>Hylobates syndactylus</i>	6	Diurnal	123.7	4.4	9.7	1.2	447.6	56.4	2.22	0.53
<i>Indri indri</i>	10	Diurnal	103.3	2.7	4.5	0.7	427.6	29.6	1.06	0.20
<i>Lagothrix lagotricha</i>	6	Diurnal	105.9	5.6	7.0	1.4	424.4	30.7	1.66	0.39
<i>Lemur catta</i>	13	Diurnal	83.5	3.8	3.2	0.5	258.9	23.8	1.24	0.19
<i>Leontopithecus chrysomelas</i>	3	Diurnal	57.3	3.4	3.1	0.2	112.5	2.2	2.78	0.15
<i>Leontopithecus rosalia</i>	7	Diurnal	55.1	0.8	2.9	0.4	110.3	7.8	2.65	0.52

Anthropoid Visual Acuity

<i>Lepilemur leucopus</i>	7	Nocturnal	50.6	1.0	2.2	0.3	146.0	7.8	1.51	0.14
<i>Lepilemur mustelinus</i>	8	Nocturnal	58.9	1.7	2.3	0.2	218.6	22.2	1.05	0.15
<i>Loris tardigradus</i>	11	Nocturnal	48.7	2.8	1.8	0.2	216.8	19.2	0.84	0.09
<i>Macaca fascicularis</i>	6	Diurnal	109.7	12.7	8.1	1.5	392.7	62.6	2.09	0.50
<i>Macaca mulatta</i>	6	Diurnal	118.7	10.7	8.8	2.1	372.5	27.5	2.39	0.65
<i>Macaca nemestrina</i>	8	Diurnal	138.2	11.7	9.0	1.7	504.2	46.3	1.80	0.34
<i>Microcebus murinus</i>	6	Nocturnal	32.8	1.3	1.4	0.2	77.6	4.5	1.86	0.33
<i>Microcebus rufus</i>	10	Nocturnal	31.1	1.8	1.2	0.8	72.4	9.6	1.68	0.38
<i>Miopithecus talapoin</i>	9	Diurnal	72.8	3.7	6.7	1.2	227.1	21.0	2.93	0.42
<i>Mirza coquereli</i>	6	Nocturnal	50.7	0.6	2.4	0.5	157.0	9.7	1.54	0.34
<i>Nasalis larvatus</i>	9	Diurnal	119.1	10.6	10.0	1.5	426.7	46.2	2.47	0.45
<i>Nycticebus coucang</i>	12	Nocturnal	57.9	1.9	2.4	0.2	239.2	16.6	1.01	0.13
<i>Nycticebus pygmaeus</i>	2	Nocturnal	52.4	0.4	2.6	0.1	212.9	23.8	1.20	0.08
<i>Otolemur crassicaudatus</i>	16	Nocturnal	64.9	4.7	3.1	0.5	252.5	29.2	1.25	0.20
<i>Otolemur garnettii</i>	7	Nocturnal	67.5	4.5	4.4	0.8	272.3	36.1	1.63	0.25
<i>Pan troglodytes</i>	10	Diurnal	187.1	9.6	14.0	2.0	899.6	147.0	1.58	0.26
<i>Perodicticus potto</i>	16	Nocturnal	61.7	2.5	2.4	0.5	193.2	13.1	1.24	0.20
<i>Phaner furcifer</i>	7	Nocturnal	52.4	2.2	2.7	0.5	166.9	11.7	1.63	0.27
<i>Pithecia pithecia</i>	9	Diurnal	75.9	3.4	5.0	0.6	215.3	17.9	2.34	0.44
<i>Pongo pygmaeus</i>	11	Diurnal	213.5	21.3	17.0	3.3	1189.0	263.0	1.44	0.27
<i>Presbytis comata</i>	8	Diurnal	92.8	2.8	8.8	0.9	396.1	22.0	2.22	0.27
<i>Presbytis frontata</i>	8	Diurnal	95.24	2.4	8.7	1.3	365.9	23.7	2.38	0.46
<i>Presbytis melalophos</i>	12	Diurnal	91.8	4.2	7.9	1.3	401.4	37.9	1.96	0.25
<i>Procolobus badius</i>	7	Diurnal	102.8	6.6	8.2	0.9	408.8	54.5	2.05	0.43
<i>Propithecus diadema</i>	7	Diurnal	89.4	4.4	4.3	0.5	392.5	40.1	1.11	0.18
<i>Propithecus verreauxi</i>	13	Diurnal	81.3	1.4	3.4	0.5	302.9	23.9	1.11	0.14
<i>Pygathrix nemaeus</i>	7	Diurnal	109.2	6.9	8.0	1.0	491.7	68.3	1.64	0.19

Table 4. Continued

Species	N	Activity	P-I length	SD	OF area	SD	Orbit area	SD	OFI	SD
<i>Pygathrix roxellana</i>	6	Diurnal	113.2	7.1	9.4	0.7	454.0	60.5	2.10	0.17
<i>Saguinus fuscicollis</i>	6	Diurnal	45.2	1.1	3.0	0.4	80.2	12.2	3.77	0.47
<i>Saguinus leucopus</i>	7	Diurnal	47.8	1.7	3.2	0.3	78.4	3.7	4.08	0.33
<i>Saguinus midas</i>	6	Diurnal	49.4	1.1	3.6	0.2	77.3	6.2	4.68	0.38
<i>Saimiri oerstedii</i>	10	Diurnal	61.1	3.1	5.4	0.8	166.5	7.1	3.22	0.46
<i>Saimiri sciureus</i>	6	Diurnal	64.1	3.4	6.1	0.4	181.9	20.8	3.37	0.35
<i>Semnopithecus entellus</i>	6	Diurnal	117.8	12.7	9.9	0.7	413.6	33.6	2.41	0.23
<i>Simias concolor</i>	6	Diurnal	98.8	5.5	6.8	0.7	371.6	33.0	1.82	0.12
<i>Tarsius bancanus</i>	14	Nocturnal	38.6	1.1	3.1	0.5	271.3	22.3	1.13	0.13
<i>Tarsius spectrum</i>	7	Nocturnal	37.1	0.7	2.5	0.3	210.8	27.1	1.20	0.06
<i>Tarsius syrichta</i>	8	Nocturnal	38.9	1.2	2.6	0.2	248.6	11.4	1.07	0.11
<i>Trachypithecus cristatus</i>	9	Diurnal	92.9	5.9	7.0	0.8	403.4	21.2	1.74	0.21
<i>Varecia variegata</i>	14	Diurnal	104.6	3.1	4.0	0.6	360.3	24.7	1.10	0.15

Notes: All values are expressed as mm for linear measurements and mm² for areas. Measurements were made using museum osteological specimens, and an attempt was made to include equal numbers of male and female specimens for sexually dimorphic species. Optic foramen area (OF area) was estimated as an elliptical area, using the greatest width of the optic foramen when viewed from an external (orbital) perspective and the maximum width of the foramen perpendicular to this chord as the two principle axes. Both dimensions were measured using a dissecting microscope fit with a calibrated micrometer reticle. Orbit area was approximated as a circular area with a diameter equal to the maximum diameter of the orbit in a parasagittal plane. N = number of specimens measured; P-I length = prosthion-inion length; SD = standard deviation; OF area = optic foramen area; OFI = optic foramen index; SD is provided when sample size is greater than two. For sample sizes of two, the difference between the two values is given in the SD column.

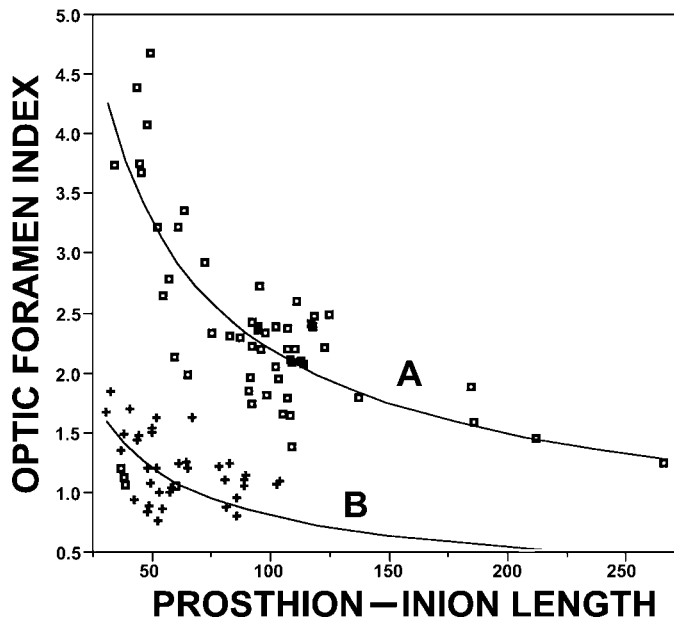


Figure 6. Bivariate plot of optic foramen index (OFI; y -axis) versus prosthion-inion length (mm; x -axis). Points are species means for the extant taxa shown in Table 4 (squares = haplorhines; crosses = strepsirrhines). Curves (shown in raw space) represent least-squares regression lines fit to $\ln$ -transformed data. Regression A was fit to non-papionin (see text) diurnal haplorhines ($\ln \text{ OFI} = 3.40 - 0.57 \times \ln \text{ P-I Length}$). Regression B was fit to nocturnal strepsirrhines ($\ln \text{ OFI} = 2.47 - 0.58 \times \ln \text{ P-I Length}$).

P-I length was chosen as a body size surrogate in this analysis because it is the linear cranial dimension most commonly preserved in the fossil specimens we have analyzed. Even in well-preserved fossil crania, the zygomatic arches and portions of the neurocranium and basicranium are often missing, making accurate estimation of cranial shape variables for size-adjustment difficult. However, it is important to keep in mind that while P-I length is highly correlated with body mass (Figure 7), these two variables are by no means equivalent. In fact, P-I length is a poor body mass surrogate in species with extreme rostral elongation. Many papionin cercopithecines, for instance, demonstrate artificially elevated OFQs relative to other species. For this reason, we have omitted the clade including *Papio*, *Mandrillus*, *Theropithecus*, *Cercocebus*, and *Lophocebus* (Disotell, 1996, 2000) from this analysis.

An additional possible source of error in this analysis is the unknown effect of ophthalmic artery size on optic foramen cross-sectional area. In primates,

Table 5. Comparison of Optic Foramen Quotients (OFQs) calculated using the diurnal anthropoid (daOFQ) and nocturnal strepsirrhine (nsOFQ) regression formulas

Species	Category	daOFQ	nsOFQ
<i>Saguinus midas</i>	DiHap	42.2	282.4
<i>Erythrocebus patas</i>	DiHap	28.3	249.9
<i>Colobus guereza</i>	DiHap	25.4	241.5
<i>Callithrix jacchus</i>	DiHap	24.5	234.2
<i>Nasalis larvatus</i>	DiHap	23.5	236.6
<i>Saguinus leucopus</i>	DiHap	21.7	227.3
<i>Homo sapiens</i>	DiHap	20.7	231.2
<i>Cercopithecus cephus</i>	DiHap	20.7	227.8
<i>Semnopithecus entellus</i>	DiHap	19.9	226.7
<i>Macaca mulatta</i>	DiHap	19.1	224.5
<i>Saimiri sciureus</i>	DiHap	18.5	219.9
<i>Hylobates syndactylus</i>	DiHap	13.2	208.6
<i>Ateles geoffroyi</i>	DiHap	12.2	205.3
<i>Miopithecus talapoin</i>	DiHap	10.8	199.7
<i>Saimiri oerstedii</i>	DiHap	10.4	197.9
<i>Chlorocebus aethiops</i>	DiHap	9.8	198.7
<i>Saguinus fuscicollis</i>	DiHap	8.8	192.2
<i>Callithrix argentata</i>	DiHap	6.4	185.7
<i>Colobus polykomos</i>	DiHap	5.8	187.9
<i>Cercopithecus petaurista</i>	DiHap	5.2	185.8
<i>Presbytis frontata</i>	DiHap	5.0	185.2
<i>Allenopithecus nigroviridis</i>	DiHap	4.8	184.8
<i>Alouatta seniculus</i>	DiHap	4.4	184.2
<i>Cebus capucinus</i>	DiHap	4.1	182.7
<i>Pan troglodytes</i>	DiHap	2.1	180.1
<i>Pygathrix roxellana</i>	DiHap	1.9	177.4
<i>Ateles fusciceps</i>	DiHap	1.8	177.3
<i>Callimico goeldii</i>	DiHap	1.3	172.6
<i>Cercopithecus mitis</i>	DiHap	0.5	173.5
<i>Pongo pygmaeus</i>	DiHap	0.2	175.5
<i>Macaca fascicularis</i>	DiHap	-0.3	171.3
<i>Gorilla gorilla</i>	DiHap	-2.3	169.7
<i>Macaca nemestrina</i>	DiHap	-2.3	166.9
<i>Hylobates klossii</i>	DiHap	-2.3	165.4
<i>Cebus albifrons</i>	DiHap	-3.1	162.9
<i>Presbytis comata</i>	DiHap	-3.5	162.1
<i>Chiropotes satanas</i>	DiHap	-5.4	156.4
<i>Procolobus badius</i>	DiHap	-5.8	156.1
<i>Cebuella pygmaea</i>	DiHap	-7.6	147.3
<i>Leontopithecus chrysomelas</i>	DiHap	-8.0	148.1
<i>Pithecia pithecia</i>	DiHap	-9.1	146.0
<i>Hylobates lar</i>	DiHap	-9.6	146.0
<i>Leontopithecus rosalia</i>	DiHap	-14.5	130.5
<i>Presbytis melalophos</i>	DiHap	-15.3	129.9
<i>Ateles paniscus</i>	DiHap	-15.3	130.4
<i>Simias concolor</i>	DiHap	-18.2	122.5
<i>Cacajao melanocephalus</i>	DiHap	-20.1	117.0
<i>Pygathrix nemaeus</i>	DiHap	-22.2	111.9
<i>Lagothrix lagotricha</i>	DiHap	-22.5	110.9
<i>Trachypithecus cristatus</i>	DiHap	-24.5	104.9

Table 5. Continued

Species	Category	daOFQ	nsOFQ
<i>Callicebus moloch</i>	DiHap	-27.6	95.2
<i>Callicebus torquatus</i>	DiHap	-29.0	91.9
<i>Alouatta palliata</i>	DiHap	-34.1	79.4
<i>Otolemur garnettii</i>	NoctStrep	-40.9	59.8
<i>Phaner furcifer</i>	NoctStrep	-48.8	37.7
<i>Varecia variegata</i>	DiStrep	-49.0	38.6
<i>Lemur catta</i>	DiStrep	-49.2	37.6
<i>Indri indri</i>	DiStrep	-51.0	33.4
<i>Eulemur fulvus</i>	CathStrep	-51.4	32.0
<i>Eulemur mongoz</i>	CathStrep	-51.9	30.4
<i>Mirza coquereli</i>	NoctStrep	-52.6	27.5
<i>Propithecus diadema</i>	DiStrep	-52.8	28.0
<i>Lepilemur leucopus</i>	NoctStrep	-53.6	25.0
<i>Galagoides zanzibaricus</i>	NoctStrep	-53.7	24.2
<i>Eulemur macaco</i>	CathStrep	-54.9	22.5
<i>Microcebus murinus</i>	NoctStrep	-55.2	19.7
<i>Propithecus verreauxi</i>	DiStrep	-55.4	20.9
<i>Otolemur crassicaudatus</i>	NoctStrep	-55.7	19.7
<i>Hapalemur griseus</i>	DiStrep	-56.9	16.4
<i>Galago matschiei</i>	NoctStrep	-57.1	15.2
<i>Perodicticus potto</i>	NoctStrep	-57.2	15.5
<i>Galago senegalensis</i>	NoctStrep	-58.9	10.3
<i>Eulemur rubriventer</i>	CathStrep	-60.2	7.9
<i>Microcebus rufus</i>	NoctStrep	-60.7	4.9
<i>Galago moholi</i>	NoctStrep	-60.8	5.2
<i>Nycticebus pygmaeus</i>	NoctStrep	-62.2	1.6
<i>Euoticus elegantulus</i>	NoctStrep	-63.6	-2.2
<i>Aotus trivirgatus</i>	NoctHap	-64.1	-3.1
<i>Hapalemur simus</i>	CathStrep	-64.4	-3.5
<i>Lepilemur mustelinus</i>	NoctStrep	-64.8	-5.0
<i>Galagoides demidoff</i>	NoctStrep	-64.9	-6.0
<i>Daubentonia</i>	NoctStrep	-66.4	-8.8
<i>madagascariensis</i>			
<i>Nycticebus coucang</i>	NoctStrep	-66.6	-9.9
<i>Galago alleni</i>	NoctStrep	-67.0	-11.2
<i>Arctocebus calabarensis</i>	NoctStrep	-67.9	-13.6
<i>Tarsius spectrum</i>	NoctHap	-69.0	-17.1
<i>Tarsius bancanus</i>	NoctHap	-70.0	-19.7
<i>Tarsius syrichta</i>	NoctHap	-71.7	-24.3
<i>Cheirogaleus major</i>	NoctStrep	-71.7	-23.9
<i>Arctocebus aureus</i>	NoctStrep	-73.0	-27.4
<i>Cheirogaleus medius</i>	NoctStrep	-73.5	-28.9
<i>Loris tardigradus</i>	NoctStrep	-74.6	-31.8
<i>Avahi laniger</i>	NoctStrep	-75.5	-33.9

Notes: Species are sorted in descending order of daOFQ (note the close correspondence with the nsOFQ sequence). For convenience, species are identified as diurnal haplorhines (DiHap), nocturnal haplorhines (NoctHap), diurnal strepsirrhines (DiStrep), cathemeral strepsirrhines (CathStrep), or nocturnal strepsirrhines (NoctStrep). Expected OFI based on the diurnal anthropoid regression was calculated using the following formula: $\ln \text{OFI} = 3.39983 - 0.56624 \times \ln \text{PI Length}$. Expected OFI based on the nocturnal strepsirrhine regression was calculated using the following formula: $\ln \text{OFI} = 2.46913 - 0.5813 \times \ln \text{PI Length}$.

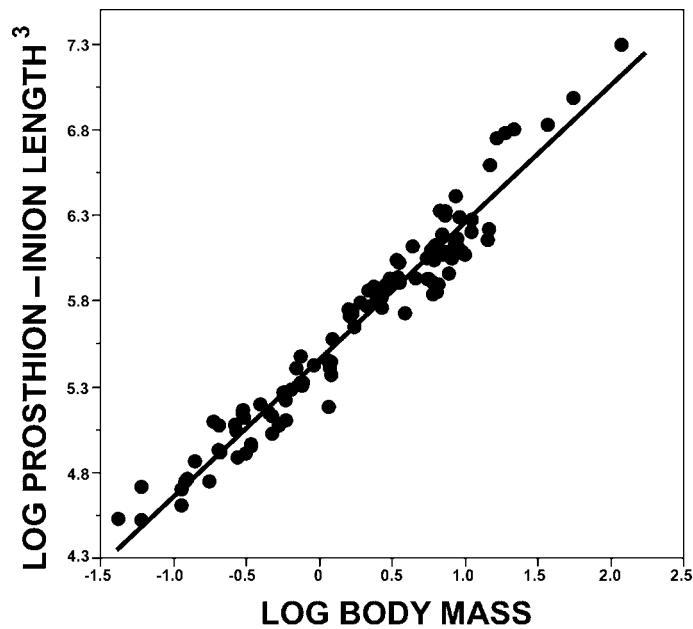


Figure 7. Bivariate plot of $\log_{10}$ prosthion-inion length³ (y -axis) versus $\log_{10}$ body mass (x -axis) in 101 primate species. Least-squares regression: $Y = 5.46 + 0.80 X$; $r^2 = 0.95$. Spearman Rho = 0.97 ($p < 0.0001$).

the ophthalmic artery supplies blood to the orbital contents and is the only major structure other than the optic nerve to traverse the optic foramen. While the ophthalmic artery occupies only a small fraction of the area of the optic canal in human anatomy, this may not be the case in all other primates. For instance, in lemurids the orbital contents are also supplied by an anastomotic branch of the ramus superior of the stapedia artery. By contrast, in loriforms the stapedia root involutes early in embryonic development, and as a result the ophthalmic artery supplies blood to the middle meningeal artery in addition to the orbital contents (Bugge, 1974). It therefore seems reasonable to expect that the ophthalmic artery will be reduced in size in lemurids and enlarged in loriforms relative to the condition seen in anthropoids. Unfortunately, however, there are no published quantitative data with which to test the effect of variation in ophthalmic artery size on optic foramen cross-sectional area.

In spite of these concerns, support for the methodology used in this analysis is provided by the fact that the optic nerve cross-sectional areas published by Stephan et al. (1984) are highly correlated with the observed optic foramen

areas listed in Table 4 (Figure 2). This high correlation is particularly impressive given the fact that optic nerve cross-sectional area may exhibit substantial intraspecific variation as the result of normal differences among individuals in the total number of optic nerve fibers (Figure 3; Jonas et al., 1992).

More importantly, calculation of OFQs for primates yields results that are very similar to two independent estimates of retinal summation (Figure 8; Kay and Kirk, 2000). These estimates include the ratio of ganglion cells to photoreceptor cells in the central retina (Figure 8B) and the ratio of optic nerve cross-sectional area to retinal area (Figure 8C). This close correspondence between estimates of retinal summation using osteological, gross morphometric, and histological measurements provides strong *a posteriori* support for the methods used in this analysis (Kay and Kirk, 2000).

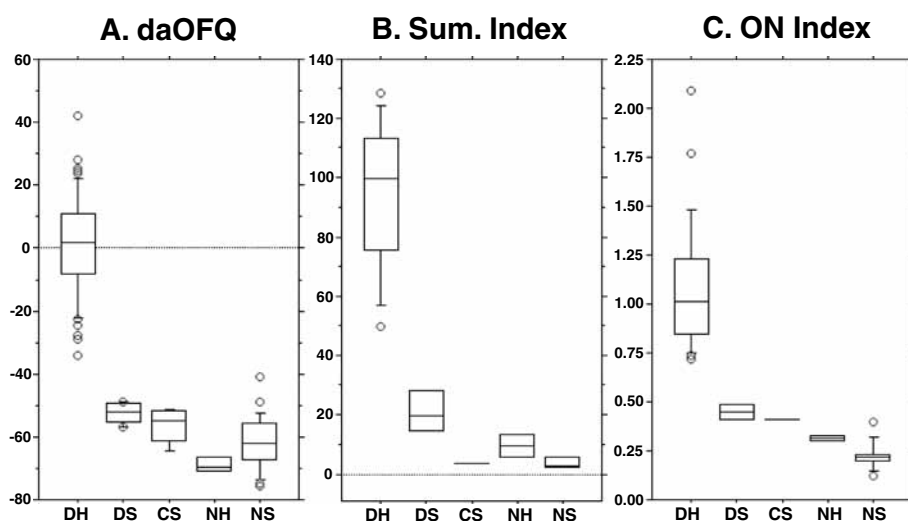


Figure 8. Quantile box plots comparing three independent estimates of retinal summation. The Optic Foramen Quotient (A) was calculated using osteological measurements according to the methods described in the text. The Summation Index (B) was calculated using histological data provided by Rohen and Castenholz (1967), and is the ratio of ganglion cells and displaced amacrine cells to photoreceptor cells in the central or perifoveal retina $\times 100$ (Kay and Kirk, 2000). The Optic Nerve Index was calculated using morphometric data provided by Stephan et al. (1984), and is the ratio of optic nerve cross-sectional area to retinal area $\times 100$ (Kay and Kirk, 2000). DH = diurnal haplorhines, DS = diurnal strepsirrhines, CS = cathemeral strepsirrhines, NH = nocturnal haplorhines, NS = nocturnal strepsirrhines.

What Does the Optic Foramen Quotient Measure?

The biological significance of the OFQ lies in the fact that optic foramen size may be used to estimate the total number of ganglion cell axons, while orbit size may be used to estimate retinal area. The size-adjusted ratio of these two variables is thus roughly equivalent to the mean density of individual sampling units (i.e., ganglion cell receptive field centers) across the retina. However, in primates, most differences in retinal anatomy are found in the central visual area rather than the much larger retinal periphery. Relative to the central visual areas, the peripheral retinas of all primates (regardless of suborder or activity pattern) have greater summation, larger ganglion cell receptive fields, are dominated by rods, and support very poor color discrimination and low visual acuity (Allman, 1977; Andrade da Costa and Hokoç, 2000; Curcio et al., 1990; DeBruyn et al., 1980; Dkhissi-Benyahya et al., 2001; Hendrickson et al., 2000; Polyak, 1957; Rohen and Castenholz, 1967; Stone and Johnston, 1981; Troilo et al., 1993; Walls, 1942; Wikler and Rakic, 1990; Wilder et al., 1996; Yamada et al., 1998, 2001). Indeed, outside of the central retina, humans and macaques have cone densities and degrees of retinal summation not very different from those of domestic cats (Goodchild et al., 1996). These factors suggest that the overwhelming majority of the variation in OFQ values seen in living primates may be explained by differences in *central* retinal anatomy rather than the organization of the entire retina.

Furthermore, even within the central retina, all primates appear to share relatively high photoreceptor densities. For instance, Wikler and Rakic (1990) found peak densities of photoreceptors in the central retinas of diurnal and nocturnal primates to be on the order of several hundreds of thousands of cells per square mm (i.e., *Macaca mulatta*, 160,000 cones mm² and 180,000 rods mm²; *Aotus trivirgatus*, 7,000 cones mm² and 325,000 rods mm²; *Otolemur garnettii*, 8,500 cones mm² and 450,000 rods mm²). While not wishing to minimize the significance of these differences (e.g., both nocturnal taxa evidently have much larger total rod populations than the diurnal *Macaca*), it is evident that the peak photoreceptor densities of diurnal anthropoids are much more similar to those of nocturnal primates than to other diurnal mammals. For example, peak retinal photoreceptor density is 68,000 cells mm² in the diurnal squirrel *Spermophilus beecheyi* and 35,000 cells mm² in the diurnal tree shrew *Tupaia belangeri* (Long and Fisher, 1983; Müller and Peichl, 1989).

Accordingly, if most variation in primate visual abilities can be attributed to differences in central retinal organization, and all primates have similar central

photoreceptor densities, it follows that central retinal summation (and thus central ganglion cell density) must be driving differences in visual acuity across primates. This conclusion is supported by the fact that diurnal anthropoids have much higher peak ganglion cell densities than other primates (Tetreault et al., this volume), although these data must be interpreted with caution because central ganglion cell densities are not directly comparable in foveate and afoveate species. Additional support is provided by the tendency of rods to summate far more extensively than cones. For example, in the central retina of *Otolemur garnettii*, summation of photoreceptors onto parvocellular ganglion cells (i.e., those ganglion cells responsible for relaying high acuity/high contrast visual information) is estimated at 307 rods per ganglion cell, but only 5 cones per ganglion cell (Yamada et al., 1998). This difference in summation between rods and cones appears to be a universal feature of retinal organization in vertebrates, and is a major factor in explaining the relatively poor acuity of rod-mediated vision (Polyak, 1957; Rodieck, 1973; Walls, 1942). Accordingly, because strepsirrhines, tarsiers, and owl monkeys have larger proportions of rods in the central visual area than anthropoids, these taxa would be expected to have higher degrees of retinal summation and lower visual acuity. These expectations are largely confirmed by the available comparative data on retinal summation (Rohen and Castenholz, 1967; Stephan et al., 1982) and visual acuity (Tables 1 and 2).

The fact that most variation in retinal anatomy within primates is confined to the central visual area helps to explain the very high correlation between OFQs and estimates of *central* retinal summation specifically (Kay and Kirk, 2000; Rohen and Castenholz, 1967). OFQs within primates may thus be viewed as a proxy for the relative degree of central retinal summation and attendant variation in visual acuity.⁵ As noted above, such differences in summation and acuity may be largely the result of differences in the relative proportion of rods to cones within the central retina. The rod/cone ratio, and thus the OFQ, is influenced by both activity pattern (e.g., as in comparisons of nocturnal, cathemeral, and diurnal strepsirrhines) and phylogeny (as in comparisons of diurnal strepsirrhines and diurnal anthropoids; Kay and Kirk, 2000).

⁵ Although OFQs are useful for making comparisons among primates, they are probably not valid for making comparisons between species with different retinal and ocular bauplans (e.g., *Macaca* vs. *Tupaia*). Indeed, if most retinal variation within primates was not limited to the central visual area, then the OFQ would be of limited interpretive utility because it could only be used to estimate average summation across the entire retina.

Optic Foramen Quotients of Extant Primates

OFQs for the 93 species of extant primates ($n=735$ total specimens) presented in Table 5 are grouped according to suborder and activity pattern in Figure 9. Statistical comparisons between these subgroups for both OFQs are presented in Table 6. Regardless of which model is used to calculate the OFQ (i.e., a nocturnal strepsirrhine model or a diurnal anthropoid model), the results are nearly identical. In each case, diurnal anthropoids have significantly higher OFQs than all other primates. These high values may primarily be attributed to the uniform presence of an all-cone, low-summation fovea in the diurnal anthropoid central retina. Similarly, diurnal strepsirrhines have significantly higher OFQs than nocturnal strepsirrhines, although these values are much lower than those of diurnal anthropoids. This disparity likewise can be explained by differences in central retinal anatomy, since diurnal strepsirrhines have better-developed areae centrales with larger proportions of cones and lower summation than do most nocturnal primates (Kay and Kirk, 2000;

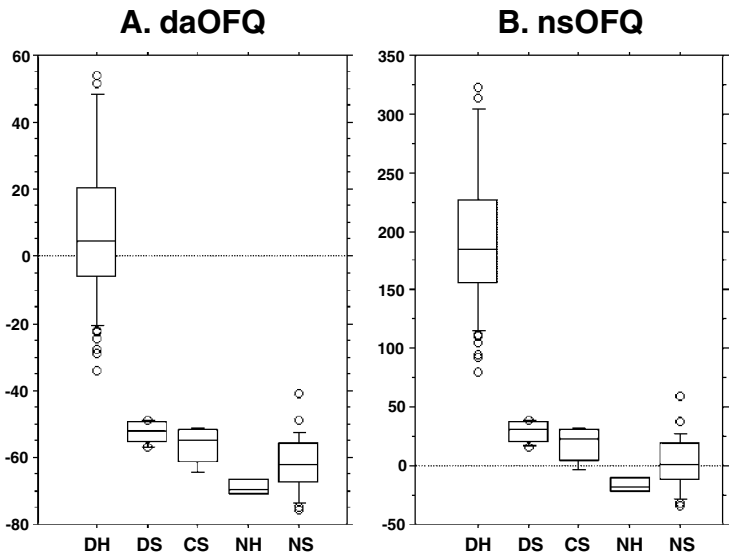


Figure 9. Quantile box plots comparing optic foramen quotients calculated using the diurnal anthropoid regression line (A, daOFQ) and the nocturnal strepsirrhine regression line (B, nsOFQ). DH = diurnal haplorhines, DS = diurnal strepsirrhines, CS = catheimeral strepsirrhines, NH = nocturnal haplorhines, NS = nocturnal strepsirrhines. Data from Table 4.

Table 6. Mann-Whitney U tests comparing the Optic Foramen Quotients (OFQs) of extant taxa grouped according to suborder and activity pattern

	Diurnal haplorhines	Diurnal strepsirrhines	Cathemeral strepsirrhines	Nocturnal haplorhines	Nocturnal strepsirrhines
Diurnal haplorhines		$U = 0.0^{***}$ $p < 0.0001$	$U = 0.0^{***}$ $p = .0002$	$U = 0.0^{***}$ $p = .0009$	$U = 0.0^{***}$ $p < .0001$
Diurnal strepsirrhines	$U = 0.0^{***}$ $p < .0001$		$U = 8.0$ NS $p = .2012$	$U = 0.0$ NS $p = .0105$	$U = 22.0^{**}$ $p = .008$
Cathemeral strepsirrhines	$U = 0.0^{***}$ $p = .0002$	$U = 8.0$ NS $p = .2012$		$U = 1.0^*$ $p = .0275$	$U = 33.0$ NS $p = .1007$
Nocturnal haplorhines	$U = 0.0^{***}$ $p = .0009$	$U = 0.0$ NS $p = .0105$	$U = 1.0^*$ $p = .0275$		$U = 26.0$ NS $p = .129$
Nocturnal strepsirrhines	$U = 0.0^{***}$ $p < .0001$	$U = 19.0^{**}$ $p = .0051$	$U = 33.0$ NS $p = .1007$	$U = 25.0$ NS $p = .1138$	

Notes: OFQs above the empty diagonal boxes were calculated using the diurnal anthropoid regression line (daOFQs). OFQs below the empty diagonal boxes were calculated using the nocturnal strepsirrhine regression line (nsOFQs). All significant results are shown in bold and identified with asterisks: * = significant at $p > .05$; ** = significant at $p > .01$; *** = significant at $p > .001$

Rohen and Castenholz, 1967; Stephan et al., 1984; Wolin and Massopust, 1970). Although cathemeral strepsirrhines show a range of OFQs that are intermediate between those of diurnal strepsirrhines and nocturnal strepsirrhines, the difference in mean OFQ between cathemeral and diurnal strepsirrhines is not statistically significant (Table 6).

Among nocturnal species, haplorhines (i.e., *Tarsius* and *Aotus*) have very low OFQs as would be expected for taxa known to have relatively high central retinal summation (Hendrickson et al., 2000; Kay and Kirk, 2000; Rohen and Castenholz, 1967; Stephan et al., 1984; Wikler and Rakic, 1990; Yamada et al., 2001). Nocturnal strepsirrhines span a much wider range of OFQs, presumably due to the variable development of an area centralis among these species (Rohen and Castenholz, 1967). In particular, *Otolemur garnettii* and *Phaner furcifer* have very large optic foramina for nocturnal species, and thus demonstrate unusually high OFQs. However, because other closely-related taxa (e.g., *Otolemur crassicaudatus*, *Microcebus rufus*) have much lower OFQs, we suspect that the high values for *O. garnettii* and *P. furcifer* are aberrant and not indicative of significantly lower summation or higher visual acuity. One possible explanation for this finding is that the ophthalmic artery is hypertrophied in these two species, producing an exceptionally large optic foramen. Unfortunately, optic nerve cross-sectional areas have not been published for

either species, so resolution of this question must await a more detailed comparison of optic nerve and optic foramen size.

These results for extant species are very similar to those published previously using a smaller sample of 45 extant primate species ($n = 320$ total specimens) that had relatively poor sampling of catarrhine diversity (Kay and Kirk, 2000). Furthermore, the current analysis demonstrates that OFQs may be calculated using empirical lines fit to very different reference groups and still yield nearly identical results (provided that the groups used have little within-group variation in retinal anatomy). Indeed, the statistical correlation between the nsOFQ and daOFQ is highly significant (Spearman Rho = 0.9997, $p < 0.0001$). This comparability of OFQ values calculated using different extant groups as a point of reference and different sample sizes demonstrates that the OFQ is a robust means of assessing central retinal summation despite the empirical basis for its calculation.

OPTIC FORAMEN QUOTIENTS IN FOSSIL PRIMATES—A PALEONTOLOGICAL RECORD OF RETINAL SUMMATION AND VISUAL ACUITY

Results of the Current Analysis

Table 7 presents the OFIs and OFQs for 16 specimens of Paleogene primates that preserve evidence of the optic foramen. Because the daOFQs and nsOFQs for fossil taxa yield nearly identical results relative to extant species, comparisons will henceforth be made with reference only to daOFQ values. Taxa represented by the fossil specimens listed in Table 7 include the adapiforms *Adapis parisiensis*, two species of *Leptadapis* (“group I” and “group II,” following Lanèque, 1993), and *Pronycticebus gaudryi*, the omomyiforms *Microchoerus*, *Necrolemur antiquus*, and *Necrolemur major*,⁶ the stem anthropoid *Simonsius* (= *Parapithecus*) *grangeri*,⁷ and two species with uncertain phylogenetic

⁶ In this analysis, *Necrolemur major* is represented by a single cranium (M 4047) currently under study by Dr Marc Godinot at the MNHN. Alpha taxonomy for this specimen follows the suggestion of Dr Godinot (personal communication). M 4047 is slightly larger and dentally more derived than QU 11060 (*N. antiquus*).

⁷ As noted by Simons and Rasmussen (1991), there is currently a lack of consensus regarding the proposed congeneric status of *Parapithecus fraasi* and *Simonsius* (= *Parapithecus*) *grangeri*. In this analysis, we have chosen to recognize the validity of *Simonsius* as a genus distinct from *Parapithecus* on the basis of the following lines of evidence: (a) Due to breakage of the mandibular symphysis in the only

Table 7. Cranial dimensions, Optic Foramen Indices (OFIs), and Optic Foramen Quotients (OFQs) of fossil specimens included in this analysis

Species	Cat. no.	P-I length	OF area	Orbit area	OFI	daOFQ	nsOFQ
<i>Adapis parisiensis</i>	MaPhQ 344	85.6	2.3	147.4	1.59	-34.2	78.6
<i>Adapis parisiensis</i>	MaPhQ 345	85.0	1.9	136.8	1.41	-41.6	58.5
<i>Adapis parisiensis</i>	ACQ 211	84.0	1.7	120.8	1.43	-41.3	59.2
<i>Adapis parisiensis</i>	MaPhQ 347	81.5	1.6	124.7	1.31	-47.2	43.0
<i>Leptadapis magnus</i> II	QU 10870	119.2	2.8	373.3	0.75	-62.3	2.8
<i>Leptadapis magnus</i> II	MaPhQ 210	123.5	3.5	271.7	1.30	-33.8	80.6
<i>Leptadapis magnus</i> II	QU 10875	106.1	3.6	376.7	0.95	-55.3	21.7
<i>Leptadapis magnus</i> I	QU 11002	108.1	3.5	317.3	1.11	-47.7	42.4
<i>Leptadapis magnus</i> I	YPM 11481	95.5	3.5	314.2	1.13	-50.3	34.9
<i>Microchoerus</i> sp.	QU 10879	46.3	1.7	141.0	1.21	-64.4	-4.4
<i>Pronycticebus gaudryi</i>	QU 11056	63.0	2.8	254.5	1.10	-61.5	3.9
<i>Rooneyia viejaensis</i> (right)	TMM 406887	50.0	2.2	120.8	1.83	-44.0	50.5
<i>Rooneyia viejaensis</i> (left)	TMM 406887	50.0	2.6	120.8	2.12	-35.2	74.3
<i>Plesiopithecus teras</i>	CGM 12393	52.0	2.5	229.7	1.07	-66.6	-10.0
<i>Necrolemur antiquus</i>	QU 11060	40.2	1.7	134.8	1.28	-65.4	-7.3
<i>Necrolemur major</i>	M 4047	43.8	2.0	153.9	1.33	-62.3	1.3
<i>Simonsius grangeri</i>	DUPC 18651	65.8	3.46	138.9	2.49	-11.0	140.4

Abbreviations and measurement protocols are the same as in Table 4. OFQs based on the diurnal anthropoid regression line (daOFQ) and the nocturnal strepsirrhine regression line (nsOFQ) were calculated as described in the text. Division of the *Leptadapis magnus* hypodigm into two unnamed species (herein designated as I and II) is based on variation in cranial anatomy and follows Lanèque (1993). P-I length could not be measured for MaPhQ 347 due to damage, so the value was estimated for this specimen using the mean P-I length of 12 more complete specimens. Measurements for *Leptadapis*, *Pronycticebus*, and *Plesiopithecus* follow Kay and Kirk, 2000. Measurements for all specimens of *Adapis*, *Rooneyia*, and *Necrolemur* are new or have been revised based upon restudy of the original specimens. Measurements for *Simonsius grangeri* were provided by Bush et al. (this volume). CGM = Cairo Geological Museum; DUPC = Duke University Primate Center; M = Swedish National Museum; MaPhQ = Montauban, Phosphorites du Quercy (collections of the Université de Montpellier); QU = Quercy (Musée National d'Histoire Naturelle, Paris); TMM = Texas Memorial Museum; YPM = Yale Peabody Museum.

Continued

known specimen of *P. fraasi*, it is uncertain how many incisors this species possessed. What is certain is that the holotype of *P. fraasi* retained at least one pair of mandibular incisors well into adulthood with no evidence of incisor root resorption. This condition contrasts with that of *S. grangeri*, in which every known adult specimen preserving the relevant morphology lacks all mandibular incisors. Even if the incisors of the type specimen of *P. fraasi* are deciduous incisors that were retained into adulthood as suggested by Kay and Simons (1983), we feel that adult retention of deciduous incisors in one species and not in the other would be sufficient to warrant generic distinction. (b) Kay and Williams (1994, pp. 412-420) have identified several dental synapomorphies that *Apidium* and *P. fraasi* share to the exclusion of *S. grangeri* (including the presence of p3-4 hypoconids and absence of a p4 paraconid) which suggest that *S. grangeri* may be the sister taxon of a *Parapithecus* + *Apidium* clade. If this phylogenetic hypothesis is correct, then the genus *Parapithecus* as constructed by Simons and Rasmussen (1991) would be paraphyletic.

affinities, *Rooneyia viejaensis* and *Plesiopithecus teras*. Measurements for *Simonsius grangeri* were kindly provided by Bush et al. (this volume), and are based on micro-CT scans of the left optic foramen. Of the fossil taxa included in this study, *Simonsius*, *Leptadapis*, *Adapis*, and *Rooneyia* are considered to have been diurnal on the basis of small relative orbit size, while *Plesiopithecus*, *Pronycticebus*, *Necrolemur*, and *Microchoerus* had relatively large orbits and were probably nocturnal (Kay and Kirk, 2000; Simons, 2001).

Figure 10 presents OFQs for presumably nocturnal fossil taxa relative to OFQs of extant groups. As expected, all of these fossil species have low OFQs that are similar to those of extant nocturnal primates, but fall well below the range of OFQs characteristic of living diurnal species. By contrast, specimens of presumably diurnal fossil taxa (Figure 11) span a wider range of OFQs. Most of these fossil crania (including all specimens of *Adapis*⁸ and *Leptadapis* group I, the QU 10875 and MaPhQ 210 specimens of *Leptadapis* group II, and the single specimen of *Rooneyia*⁹) have OFQs that fall within or slightly above the range of OFQs for extant diurnal strepsirrhines. However, the QU 10870 specimen of *Leptadapis* group II has a daOFQ (−62.3) that is slightly lower than the species mean of *Hapalemur griseus*, which exhibits the lowest daOFQ (−56.9) of any living diurnal strepsirrhine. It is currently unclear whether this low OFQ for QU 10870 is the result of normal variation within

⁸ A previous analysis of the OFQ of *Adapis parisiensis* suggested that this species had an extremely low OFQ despite the fact that it was presumably diurnal on the basis of orbit diameter (Kay and Kirk, 2000). However, this judgement was based on the measurements of a single specimen (MaPhQ 345). Restudy of this specimen, and comparison with three additional specimens of *A. parisiensis* that have much better-preserved optic foramina, suggest that our previously published estimate of optic foramen area for MaPhQ 345 was too low. The measurable optic foramen in this specimen is poorly preserved due to chemical weathering of the foramen margin, incomplete preparation of the specimen, and damage to the bony pillar separating the optic foramen and superior orbital fissure. Additional comparisons with better-preserved *A. parisiensis* specimens have permitted us to remeasure the major axis of the optic foramen in MaPhQ 345 with a higher degree of confidence. The new optic foramen area provides revised OFQs for this specimen that are very close to the mean for the species (Table 7).

⁹ In contrast to our previous analysis optic foramen size (Kay and Kirk, 2000), reexamination of the type specimen of *Rooneyia viejaensis* (TMM 406887) has led us to include OFQs calculated using the dimensions of both the right and left optic foramina. In TMM 406887, the pillar of bone separating the optic foramen and the inferior orbital fissure of the right orbit is broken and slightly deviated superomedially, suggesting that the right optic foramen area may be underestimated based on the measurements presented here. By contrast, the pillar separating the left optic foramen and inferior orbital fissure shows no sign of breakage, but the matrix in the apex of the left orbit has been less-completely removed during preparation. These observations suggest that the true medial edge of the left optic foramen may be slightly lateral to the currently visible edge, and that the left optic foramen area presented here may be an overestimate. For the purposes of this analysis, we take our measurements of the left and right optic foramina of TMM 406887 as upper and lower limits (respectively) on the true optic foramen area of this specimen, which is probably intermediate between the two values reported here.

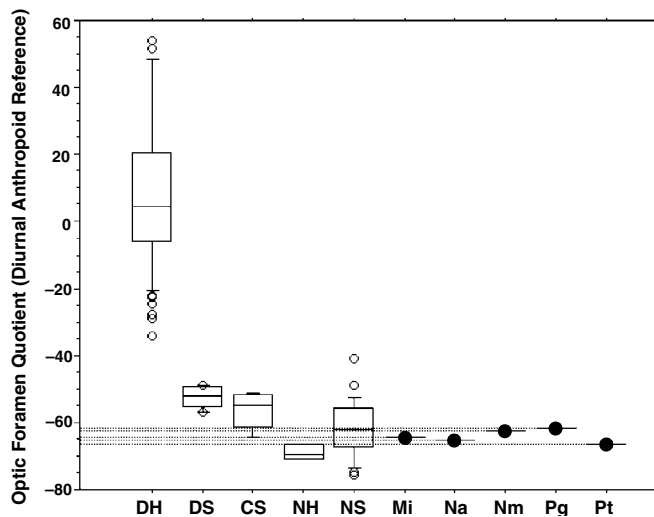


Figure 10. Quantile box plots comparing the daOFQs of extant primates and fossil primates judged to be nocturnal on the basis of orbit diameter. DH = diurnal haplorhines, DS = diurnal strepsirrhines, CS = cathemeral strepsirrhines, NH = nocturnal haplorhines, NS = nocturnal strepsirrhines, Mi = *Microchoerus* sp., Na = *Necrolemur antiquus*, Nm = *Necrolemur major*, Pg = *Pronycticebus gaudryi*, Pt = *Plesiopithecus teras*. Data from Tables 4 and 6.

the genus *Leptadapis*, measurement error, the use of P-I length as a variable for size adjustment, or is simply an artifact of poor preservation. Regardless, Figure 11 shows that the mean OFQ value for *Leptadapis* group II is within the range of living diurnal strepsirrhines.

The only fossil species with OFQs that deviate substantially from extant strepsirrhines of similar activity pattern is the early Oligocene stem anthropoid *Simonsius grangeri*. The single known specimen of *S. grangeri* that preserves evidence of the optic foramen (DUPC 18651) has a very high daOFQ comparable to those of diurnal crown anthropoids, and well above the range of all living strepsirrhines (Figure 11). While the type specimen of *Rooneyia* and one specimen each of *Adapis* (MaPhQ 344) and *Leptadapis* group II (MaPhQ 210) also demonstrate daOFQs that fall above the range of all extant strepsirrhine species means, the mean daOFQs for all three taxa fall below the lower end of the extant diurnal haplorhine range (Figure 11). *S. grangeri* is thus unique among the fossil species included in this analysis in exhibiting very high OFQs similar to those of living diurnal haplorhines.

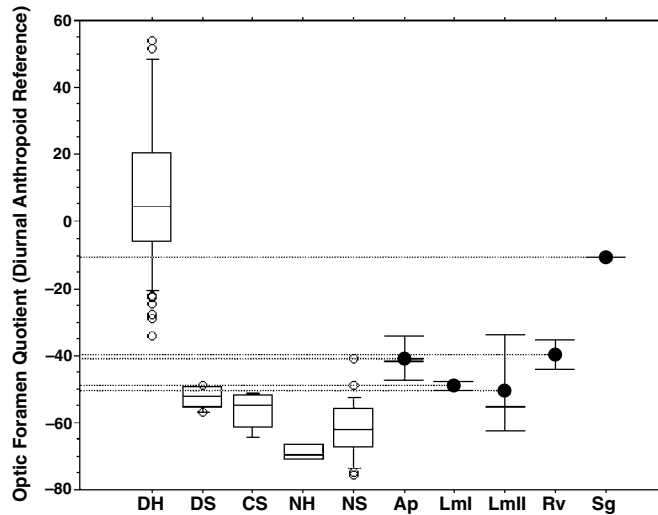


Figure 11. Quantile box plots comparing the daOFQs of extant primates and fossil primates judged to be diurnal on the basis of orbit diameter. DH = diurnal haplorhines, DS = diurnal strepsirrhines, CS = cathemeral strepsirrhines, NH = nocturnal haplorhines, NS = nocturnal strepsirrhines, Ap = *Adapis parisiensis*, LmI = *Leptadapis magnus* (I), LmII = *Leptadapis magnus* (II), Rv = *Rooneyia viejaensis*, Sg = *Simonsius grangeri*. For fossil taxa, vertical lines represent the range of values calculated for each species, with the filled circle and dotted horizontal line representing the mean. Solid horizontal lines represent individual specimens (*Adapis*, *Leptadapis*, *Simonsius*) or optic canals (*Rooneyia*). Data from Tables 4 and 6.

Implications for Anthropoid Origins—Non-Anthropoid Fossil Taxa

The results for the non-anthropoid fossil taxa included in this analysis suggest that these species exhibited degrees of central retinal summation, central rod:cone ratios, and visual acuity broadly comparable to extant strepsirrhines of similar activity pattern. Accordingly, *Plesiopithecus*, *Pronycticebus*, *Necrolemur*, and *Microchoerus* are inferred to have had central retinas with large proportions of high-summation rods. Visual acuity in these nocturnal genera was probably low relative to other primates (ca. 5–6 c/deg.), as in the living nocturnal strepsirrhines *Microcebus* and *Otolemur* (Tables 1 and 2). By contrast, *Rooneyia*, *Adapis*, and both species of *Leptadapis* are inferred to have exhibited well-developed areae centrales with larger proportions of cones and lower degrees of summation than the nocturnal fossil species. Because the mean OFQs of both species of *Leptadapis* fall within the range of OFQs

reported for extant diurnal strepsirrhines (Tables 5 and 7), the visual acuity of *Leptadapis* was probably similar to that of the diurnal strepsirrhine *Lemur catta* (6–7 c/deg.; Neuringer et al., 1981). However, *Rooneyia* and *Adapis* demonstrate mean OFQs above the range of extant diurnal strepsirrhines, suggesting that these diurnal fossil taxa may have exhibited somewhat higher visual acuity than any living strepsirrhine.

The most important aspect of these results with respect to the origins of anthropoid high visual acuity is the substantial dissimilarity in OFQ values between Eocene/Oligocene prosimian-grade taxa and living diurnal anthropoids. Regardless of activity pattern or taxonomic affinity, all of the extinct non-anthropoid species included in this analysis have mean OFQs that fall well below the range of extant diurnal anthropoids. Of particular significance is the fact that even the presumably diurnal genera *Leptadapis*, *Adapis*, and *Rooneyia* have relatively low OFQs. By analogy with extant primates, our data therefore suggest that it is very unlikely that any of the fossil non-anthropoid taxa listed in Table 7 possessed all-cone foveae, low central summation, and very high visual acuity comparable to that seen in extant diurnal anthropoids.

These results have several important implications for the ancestral condition of the stem anthropoid and haplorhine clades (Figure 12). First, if anthropoids are more closely related to adapiforms than to tarsiers (Figure 12A; Franzen, 1994; Gingerich, 1980; Rasmussen, 1990), then the last common ancestor of the adapiform–anthropoid clade would likely have possessed a moderate-summation, relatively low-acuity retina like those inferred for the diurnal adapiforms *Adapis* and *Leptadapis*. This scenario would therefore imply that the foveal architecture of *Tarsius* was not inherited (and subsequently modified) from a foveate diurnal common ancestor with the anthropoid + adapiform clade. Convergent acquisition of a fovea by tarsiers and anthropoids, while possible, is rendered less likely by the fact that crown haplorhines (except *Aotus*) are unique among mammals in consistently exhibiting this character.

However, if some omomyiforms (possibly a member of the Washakiinae or Necrolemurinae) are more closely related to tarsiers than to anthropoids (Figure 12B; Beard and MacPhee, 1994; Beard et al., 1991; Rosenberger, 1985), then high-summation, low-acuity retinas like those inferred for *Microchoerus* and *Necrolemur* may also have characterized the last common ancestor of tarsiers and anthropoids. As with the adapiform–anthropoid hypothesis (Figure 12A), this scenario would suggest that derived retinal

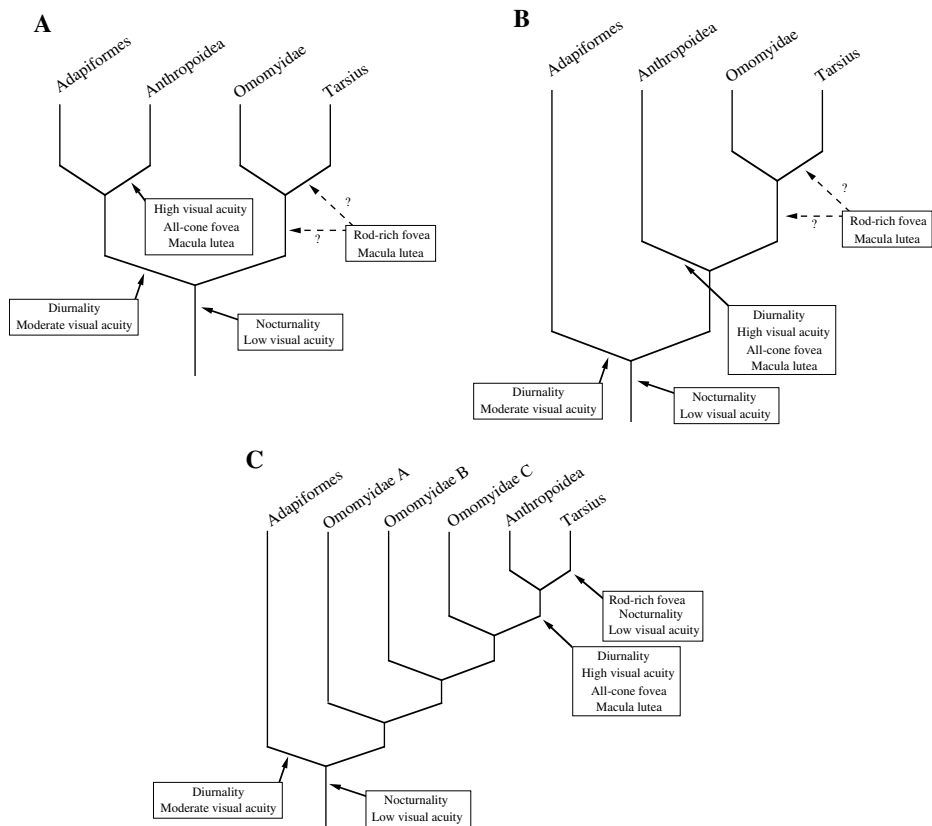


Figure 12. Simplified schematic representing alternate hypotheses of anthropoid phylogenetic relationships. (A) Adapiformes and anthropoids form a monophyletic clade; (B) Omomyiforms and *Tarsius* form a clade with anthropoids as the sister group; (C) Anthropoids and *Tarsius* form a clade with omomyiforms as the paraphyletic sister group. All phylogenetic scenarios assume that the last common ancestor of the groups in question was nocturnal and had relatively low visual acuity (Cartmill, 1970; Heesy and Ross, 2001; Martin, 1990).

similarities shared by tarsiers and anthropoids (e.g., the fovea and macula lutea) are either convergent, or that nocturnal omomyiforms (like tarsiers) retained a fovea and macula lutea from a diurnal common ancestor. As noted previously, the former possibility seems unlikely because the fovea and macula lutea are unique to crown haplorhines among mammals, while the latter possibility has been questioned on the basis of orbit size in omomyiforms (Cartmill, 1980). Most omomyiforms do not exhibit the enormously enlarged orbits (and thus eyes) characteristic of both extant tarsiers and owl monkeys,

which are presumably adaptations for increased visual sensitivity in the absence of a tapetum lucidum (Cartmill, 1980; Kay and Kirk, 2000). This lack of extreme orbital hypertrophy in many nocturnal omomyiforms may be evidence that these taxa had tapeta, unlike extant haplorhines. Comparative data on vertebrate visual systems have further shown that tapeta are almost never associated with foveae (Ross, this volume), suggesting that most omomyiforms lacked the fovea/macula lutea complex if they did possess tapeta. However, Beard et al. (1991) have shown that at least one omomyiform (*Shoshonius cooperi*) did exhibit orbital hypertrophy that may be indicative of a close relationship to living tarsiers. Additionally, Heesy and Ross (2001) have reconstructed orbital diameters for *Omomys carteri* based on partial orbital rims of fossil maxillae which suggest that this species had the largest relative orbit size of any mammal (including tarsiers). These findings are consistent with the hypothesis that at least some omomyiforms did not have tapeta, and thus may fall within the crown haplorhine clade.

Finally, if omomyiforms are a paraphyletic stem clade from which the crown Haplorhini arose (Figure 12C; Cartmill and Kay, 1978; Kay and Williams, 1994; MacPhee and Cartmill, 1986; Ross et al., 1998), then the last common ancestor of omomyiforms and crown haplorhines was probably nocturnal and possessed a high summation, low-acuity central retina. In this case, the origin of the crown Haplorhini would have been marked by a transition to a diurnal activity pattern and the appearance of an all-cone fovea and macula lutea. The fovea would subsequently have been modified in *Tarsius* through a return to a nocturnal habitus (Cartmill, 1980; Le Gros Clark, 1959; Martin, 1990; Ross, 1996, 2000).

This hypothesis of a sister-group relationship between tarsiers and anthropoids (Figure 12C) is the only phylogenetic scenario presented in Figure 12 that does not favor the convergent acquisition of the fovea/macula lutea complex in tarsiers and anthropoids. While it is not possible to definitively reject the hypothesis that the fovea and macula lutea of *Tarsius* evolved in a nocturnal setting and that a tarsier-like retinal morphology represents the primitive condition for crown haplorhines, at least three lines of comparative evidence suggest that this scenario is unlikely. Firstly, the fovea and macula lutea are, by their very nature, adaptations for enhancing visual acuity (Polyak, 1957; Ross, this volume; Walls, 1942). However, it is improbable that the visual benefits conferred by the fovea and macula lutea (e.g., elimination of minute refractive errors caused by non-photoreceptor cells in the inner retina, reduction of

chromatic aberration, etc.; Rodieck, 1988; Weale, 1966; Wolin and Massopust, 1970) could be enjoyed by tarsiers given their low foveal cone densities, low central ganglion cell densities, and high central summation relative to diurnal anthropoids (Castenholz, 1984; Hendrickson et al., 2000; Rohen and Castenholz, 1967; Tetreault et al., this volume). Ross (this volume) has also demonstrated that the fovea of *Tarsius* does not contribute measurably to visual acuity through magnification of the central retinal image. This lack of an obvious function favors the hypothesis that the tarsier fovea and macula lutea are phylogenetic “baggage” inherited from diurnal ancestors that exhibited foveae more typical of extant diurnal anthropoids (Cartmill, 1980).

Second, as an adaptation for reducing chromatic aberration and protecting the fovea from oxidative damage, the macula lutea is functionally relevant only in a diurnal setting and potentially detrimental in a nocturnal setting. Nocturnal mammals, which are limited by low environmental light levels, typically cannot afford to screen out any available photons because doing so would inevitably reduce the sensitivity of the eye (Douglas and Marshall, 1999). Furthermore, given the substantially reduced intensities of moon and star light relative to sunlight (i.e., a difference of approximately 6 log units; Lythgoe, 1979; Stair and Johnston, 1953), it is unlikely that nocturnal species would experience any selective pressure to protect their retinas from photo-oxidation. Given the fact that mammalian rod photopigments absorb light maximally between 400 nm and 600 nm in wavelength (λ_{max} B500 nm; Bowmaker, 1991; Bowmaker et al., 1991) while macular pigments strongly absorb light below 500 nm (Nussbaum et al., 1981), one would expect any rods that fall within the macula lutea to have substantially reduced sensitivity. It is therefore very difficult to reconcile the existence of a macula lutea surrounding a rod-rich fovea in tarsiers with their strongly nocturnal habits (Niemitz, 1984). Although it is unclear why tarsiers have not paralleled owl monkeys in the loss of the macula lutea,¹⁰ these functional considerations also

¹⁰ It is possible that the spectral content of light available to *Aotus* and *Tarsius* is not comparable because these genera prefer different forest strata. Perhaps because of a lack of competition from other nocturnal primates, owl monkeys range from near the ground (7 m) to the top of the canopy (35 m; Wright, 1981). By contrast, the great majority of a tarsier's activity (including hunting and traveling) occurs within 3 m of the ground (MacKinnon and MacKinnon, 1980; Niemitz, 1984). While sunlight and moonlight have approximately equal proportions of light between 400 nm and 500 nm in wavelength (Stair and Johnston, 1953), Endler (1993) has shown that filtration of sunlight by foliage causes forest shade to be largely dominated by light greater than 500 nm in wavelength. If this finding also applies to moon and starlight, then the macula lutea may not significantly impair rod function

strongly suggest that the faint macula lutea of *Tarsius* is, like the fovea it surrounds, a non-functional plesiomorphic retention.

Third, Ross has provided abundant comparative and experimental evidence (Ross, 1993, 1995, 1996, 2000; Ross and Hylander, 2000) that the postorbital septum (found in both tarsiers and anthropoids) evolved to insulate the eyes of stem haplorhines from oscillations caused by the temporalis muscles during chewing. The comparatively minute loss of acuity incurred by such masticatory oscillations would be most significant for species specialized for extremely high visual acuity like extant diurnal anthropoids. Ross' hypothesis thus suggests that postorbital septa evolved in concert with diurnality and high acuity foveae. The broad postorbital plate shared by living tarsiers (Cartmill, 1994) is therefore an additional line of evidence that diurnality and all-cone foveae are primitive for the crown haplorhine clade.

Implications for Anthropoid Origins—*Simonsius grangeri*

The fact that *Simonsius grangeri* has relatively high OFQs comparable to those of extant diurnal anthropoids is of particular significance for anthropoid origins given the undisputed status of this species as a diurnal stem anthropoid (Bush et al., this volume; Ross et al., 1998; Simons, 2001, this volume). By analogy with living primates, the high OFQs of *S. grangeri* provide strong evidence that this species exhibited the full suite of retinal adaptations related to high visual acuity that are shared by living diurnal anthropoids. *S. grangeri* is thus the only fossil taxon included in this analysis that is inferred to have possessed a low-summation all-cone fovea in the central retina. Accordingly, the visual acuity of *S. grangeri* was probably very high (ca. 40–50 c/deg), as in living diurnal anthropoids of similar body size (Bush et al., this volume). These data also suggest that the complex of traits associated with high acuity in living anthropoids is of great antiquity, extending at least to the last common ancestor of the parapithecoids and crown anthropoids. As indicated by the occurrence of several species of qatraniine parapithecoids at Quarry L-41 in the Fayum, these clades diverged no later than 36–35 million years ago, and

Continued

in tarsiers because macular pigments primarily absorb light below 500 nm. However, owl monkeys would presumably have experienced selection to lose the macula lutea because they frequently occupy upper canopy levels that would not be similarly impoverished in 400 nm–500 nm light.

possibly substantially earlier (Kappelman, 1992; Ross et al., 1998; Simons and Rasmussen, 1994; Simons et al., 2001).

While these results do not resolve the debate regarding the primitive condition of the visual system in the crown Haplorhini, the high OFQs of *S. grangeri* are at least consistent with the hypothesis that the all-cone fovea is a crown haplorhine synapomorphy that was subsequently lost in tarsiers. If, by contrast, *S. grangeri* had been shown not to possess high OFQs like extant diurnal anthropoids, then parallel acquisition of foveae in tarsiers and anthropoids would have been a much more plausible scenario. In this context, the relative size of the optic foramen in more primitive stem anthropoids such as *Eosimias* and *Bahinia* may prove useful in resolving the higher-level phylogenetic relationships of the Haplorhini.

The relatively high OFQs of *S. grangeri* are also relevant for the hypothesis that the postorbital septum of anthropoids evolved to isolate the orbital contents from perturbation by the anterior temporalis (Cartmill, 1980; Ross, 1993, 1995, 1996, 2000; Ross and Hylander, 2000). As noted previously, such oscillations are probably only significant for species (such as diurnal anthropoids) that have very high visual acuity. Simons (2001, this volume) has described the postorbital septum of *S. grangeri* as essentially indistinguishable from that of a modern anthropoid. Combined with the results of Bush et al. (this volume), the findings for *S. grangeri* and the other fossil taxa included in this analysis demonstrate that, where the necessary evidence is available, extremely high visual acuity in primates is always associated with the presence of a postorbital septum. This association provides further support for the hypothesis that eyes adapted for high visual acuity and postorbital septa evolved together as a single trait complex in either the stem Anthropoidea or Haplorhini (Cartmill, 1980; Ross, 1996).

CONCLUSIONS

1. The nocturnal Eocene/Oligocene non-anthropoid primates *Plesiopithecus*, *Pronycticebus*, *Necrolemur*, and *Microchoerus* probably had central retinas with a large proportion of high summation rods. Visual acuity in these genera was likely comparable to that of extant nocturnal primates.
2. The diurnal Eocene/Oligocene non-anthropoid primates *Rooneyia*, *Adapis*, and *Leptadapis* probably had central retinas with a larger proportion

of low summation cones. Visual acuity in these genera was likely comparable to or slightly higher than that of extant diurnal strepsirrhines.

3. The early Oligocene stem anthropoid *Simonsius* (= *Parapithecus*) *grangeri* probably resembled living anthropoids in possessing a low-summation, all-cone fovea and in exhibiting very high visual acuity.
4. The fovea/macula lutea complex that is unique to tarsiers and anthropoids among vertebrates probably evolved only once in the haplorhine stem lineage. Accordingly, the last common ancestor of crown haplorhines was probably diurnal, possessed an all-cone fovea, and had visual acuity comparable to living diurnal anthropoids. Because *Rooneyia*, the omomyiforms *Necrolemur* and *Microchoerus*, and the adapiforms *Pronycticebus*, *Adapis*, and *Leptadapis* lack evidence of either an all-cone fovea (as assessed by relative optic foramen size) or tarsier-like orbital hypertrophy, the phylogenetic affinities of these genera probably lie outside of the crown Haplorhini.

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